RegTech KESONIA: Worked Examples, Symbols and Units

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This casebook combines the original benchmark, pricing and funding cases with their symbol reference, dated contract events, options and collateral, liquidity and capital components, hedge accounting, connected finance, the implementation register and SokoIntel candidate map. Symbols and units are defined beside the equations and in Part III. Equations are numbered by part. The numerical edition remains 1 October 2026; this TeX typesetting revision was produced on 16 October 2026.

Part I. Original worked examples

28 September 2026 numerical baseline; 1 October 2026 notation and PDF edition. These examples implement useful parts of the introduced reviews. One case checks historical CBK observations. All portfolios, instrument quotes, holidays, behaviour and transactions in the remaining cases are synthetic. Read the scope register before applying a formula to a product.

The symbol and unit reference defines all mathematical notation, rate conventions, money scales and abbreviations. The standalone PDF includes this reference, the detailed operations methodology and the October 1 extensions listed below.

Reproduce

From the repository root, with Python 3.10 or later:

```
python -m unittest discover -s tests -v

python -m research_lab.run_examples
```

No third-party Python dependencies are required. The workflow case creates and removes a temporary SQLite database. The runner writes checked_examples.json, including hashes of the executing modules. It does not fetch quotes, connect to a bank or transmit instructions. Decimal is used for benchmark and monetary ledgers; curve/OLS/root-solving examples use floating point with stated residual checks.

1. Compounding is a dated observation contract

For constant principal P (KES), observation-interval index i, annual decimal benchmark fixing r_i, calendar-day weight n_i and day-count denominator B (365), define:

$$G = \prod_i (1 + r_i n_i / B), \qquad A = G - 1, \tag{1.1}$$

$$I_{\text{benchmark}} = PA, \qquad I_{\text{margin}} = Pk\,d/B. \tag{1.2}$$

Symbols: G is the dimensionless gross accumulation factor; A = G - 1 is the interest-only accrual factor. $I_{\text{benchmark}}$ and I_{margin} are interest amounts in KES. The product operator multiplies all observation-interval factors. The annual simple margin is k; d is the actual contractual accrual-day count. Observation weights n_i and contractual days d are separately retained when observation intervals shift.

The implemented profile compounds the benchmark and adds a **simple margin** over actual accrual days d. It has constant principal, no caps/floors, no in-period margin changes and no capitalisation of unpaid interest. Those limits are deliberate. The profile cannot silently substitute for another executed contract. Use `gross_accumulation_factor` and `benchmark_accrual_factor` in shared schemas: the symbol F has different meanings in the existing project manuscripts.

Historical weekend cross-check

The CBK fixing and index sources supply the following inputs:

Input/output	Value
Accrual interval	5–8 September 2025, half-open

Input/output	Value
Friday fixing, annual	9.4618%
Principal	KES 1,000,000
Benchmark weight	3 calendar days
Simple illustrative margin	3%
Benchmark interest	KES 777.68219178
Margin interest	KES 246.57534247
Total, rounded once half-up	**KES 1,024.26**
Start/end published index	100.10451025 / 100.18235974
Index-based less fixing-based interest	Approximately KES -0.00004933

The Friday observation carries one three-day **linear factor**. Replacing it with three independently compounded one-day factors changes the result. Published-index rounding explains why exact equality is not required: the test permits less than KES 0.0001 on this principal. This is an independent historical cross-check, not a full replication of CBK's unpublished precision pipeline.

Mapping, availability and corrections

The synthetic September 10–17 fixture invents a September 15 holiday. Five-business-day lookback without observation shift has weights totalling seven calendar days; shifted observation intervals have weights totalling six calendar days. Both schedules contain four observation intervals: their day weights are [1, 1, 4, 1] and [1, 1, 3, 1], respectively. This demonstrates why the schedule, weight convention and annualisation denominator must be recorded. The final CBK wording is five days; this synthetic fixture does not settle its operational interpretation.

Fixing selection is as-of a timezone-aware calculation cutoff. Later corrections cannot enter an earlier replay. Missing observations fail rather than carrying an old rate automatically. An explicitly authorised CBR fallback is possible; an available KESONIA observation takes priority. Duplicate latest versions, invalid factors, gaps and overlaps fail. Negative rates are allowed where their factors remain positive.

Implementation: benchmark.py. A production profile still needs the official holiday source, approved mapping, fallback clause, balance events, spread convention, floors/caps, correction policy and customer-statement rounding.

2. Build a curve from instruments and reprice it

Use invented unit-price annual instruments: a one-year deposit at 10%, a two-year par coupon at 12%, and a three-year par coupon at 13%. With annual unit accruals:

$$D_1 = (1 + q_1)^{-1}, \qquad D_n = \frac{1 - c_n \sum_{j=1}^{n-1} D_j}{1 + c_n}. \tag{1.3}$$

Symbols: D_j is the dimensionless value today of one unit paid at year j; D_n is the factor at the final annual node n. q_1 is the one-year annual decimal deposit quote; c_n is the annual par coupon quote of the instrument maturing at year n. The sum adds earlier annual discount factors. The price residual below is dimensionless price minus one.

Tenor	Discount factor	Instrument repricing error
1 year	0.909090909091	0
2 years	0.795454545455	about -2.22×10^{-16}
3 years	0.688857602574	0

Between nodes, interpolate log discount factors. Thus $D(1.5) = \sqrt{D(1)D(2)} = 0.850376678812$. This ensures positive factors. It does not require decreasing factors when rates are negative. Extrapolation fails without a separately supplied policy.

Projection and discounting remain separate inputs. For coupon dates t_j:

$$f_j = \frac{D_P(t_{j-1})/D_P(t_j) - 1}{t_j - t_{j-1}}, \tag{1.4}$$

$$\text{PV} = N \left[\sum_j f_j (t_j - t_{j-1}) D_D(t_j) + D_D(T) \right]. \tag{1.5}$$

Symbols: j indexes coupons; t_j is payment time in years from valuation, with t_0 = 0; T is the last payment time. D_P is the projection-curve factor representation and D_D is the discounting-curve factor. Their subscripts mean projection and discounting. f_j is the annual simple forward rate for that coupon interval; N is principal in currency; PV is present value in currency. The interval t_j - t_(j-1) is in years.

Using the same curve for both purposes gives a par floating note of **1,000,000**, by telescoping. Using a separately bootstrapped projection curve with quotes 11%, 13%, 14% gives **1,024,330.90** under the selected discount curve. The difference is an imposed basis, not an estimated Kenyan basis.

For fixed cash flows of 100,000, 100,000 and 1,100,000, central finite-difference node sensitivities are approximately **9.090909, 15.909091 and 227.323012 currency per basis point**. Tests halve the bump and compare the aggregate with the independent analytic parallel-shift derivative. This is sensitivity to specified zero-rate nodes, not regulatory market-risk capital.

Implementation: curves.py. Instrument calendars, market quotes, liquidity/basis calibration and option models remain future work.

3. Derive the margin from money

Let H be the expected collectible interest-bearing balance integrated over year fractions, in currency-years. Solve the pricing bridge before presenting a rate:

$$k = \frac{C_{\text{fund}} + C_{op} + EL_{\text{principal}} + C_{\text{cap}} - \text{Fee} - I_{\text{benchmark}}}{H}. \tag{1.6}$$

Symbols: C_{fund} is external funding expense; C_{op} is operating cost; $EL_{\text{principal}}$ is expected principal loss; C_{cap} is the required capital-return amount; Fee is included fee income; and $I_{\text{benchmark}}$ is collectible benchmark income. All are money for the same entity and horizon. The solved k is an annual decimal margin.

Capital EC is a currency stock. With annual hurdle h and constant capital over τ years, C_{cap} = h times EC times tau. This example spans one year, so C_{cap} = 0.20 times 50,000 = 10,000. The implementation accepts C_{cap} in money directly. Dated capital changes require a dated capital-return calculation. Expected loss here

is principal loss, with the income convention explicit. A model that already removes defaulted interest from expected income must not charge it again inside another loss term.

Item	Value and unit
H	1,000,000 KES-years
Benchmark income / fees	90,000 / 5,000
Funding / operating cost	100,000 / 20,000
Expected principal loss	15,000
Economic capital / annual hurdle	50,000 / 20%
Required margin income	50,000

The margin is **5%**. Total income including fees is 145,000; return before the capital hurdle is 10,000; one-year pre-tax RAROC is **20%**; surplus after the hurdle is **zero**. These outputs reconcile to one economic case, rather than adding a percentage capital stock to an interest rate.

Here RAROC means risk-adjusted return on capital: the one-year pre-tax return before the capital hurdle divided by EC. In the following separate equation, y denotes annual effective cash-flow yield.

A separate fee example advances net 990,000 and receives 1,140,000 exactly one year later. The annual cash-flow yield solves 990,000 = 1,140,000/(1+y), giving **15.151515%**. The numerical solver's PV residual is below 10^{-8} currency. This is not labelled statutory APR: the legally required included costs and conventions must first be established.

Implementation: pricing.py. The IRR helper permits one initial negative cash flow followed by nonnegative receipts, ensuring monotonicity and avoiding ambiguous multiple-root cases. It rejects other sign patterns.

4. Dated FTP, deposit estimation and replication

The operations methodology provides input calendars, equations, panels and complete tables.

H_D is drawn balance times actual interval days / 365; H_U is the unused limit times that same year fraction. Both are currency-years. Across multiple intervals, sum those interval exposures. FTP means funds transfer pricing and EAD means exposure at default, in currency. FTP margins below are money contributions over the year; pricing k above is an annual rate.

For drawn exposure-time H_D and undrawn commitment-time H_U, customer income, internal transfer charges and external funding are separate records. The business unit's income less FTP plus Treasury's FTP less external cost equals consolidated income less external cost. Conditional-default EAD does not become the interest-bearing balance.

Synthetic 2027 product	Consolidated contribution before credit/operating/capital costs
Fixed bullet	70,000.00
Amortising	43,582.19
Floating	60,000.00
Revolver, realised example path	41,379.45
Same revolver, frozen plan	37,000.00

RMSE means root mean squared error, measured here in basis points. One basis point (bp) is 0.01 percentage point or 0.0001 in decimal-rate units. A deposit beta is a dimensionless paid-rate response per unit market-rate change.

The deposit experiment estimates immediate and lagged upward/downward responses from 23 training changes per segment, then tests a frozen fit on 13 later changes. Synthetic regime change raises RMSE from approximately **0.121 to 4.209 basis points**. The experiment illustrates temporal instability; it is not an empirical performance claim.

Withdrawal, same-bank migration, new inflows and original-cohort retention are separate. A discrete ladder uses observed synthetic retention bounds, a minimum overnight allocation and a maturity limit. At the stated quotes/policy, the solution is 20% overnight, 60% six-month and 20% twelve-month, with **5.3% annualised yield**. Tightening the retention policy changes the solution and reduces yield to **5.2%**. A repricing beta is never converted directly into deposit duration.

5. Carry cash and equity across three dates

The synthetic opening bank has KES millions: cash 200, loans 800, deposits 900 and equity 100. All rates apply to opening balances with ACT/365; repayments, drawings, charge-offs, recoveries, distributions and collateral posting occur at period end. Required cash is 60; an explicitly assumed replacement facility has a total capacity of 150. Newly drawn funding incurs interest in the next period.

Period end	NII	Closing cash	Closing loans	Deposits	Wholesale debt	Equity
1 February 2026	5.138356	172.138356	710	810	0	77.138356
1 March 2026	3.706822	60	660	630	31.154822	71.845178
1 April 2026	3.314280	60	625	550	85.840543	62.159457

Collateral receivable grows from zero to 5, then 13. For each date:

$$\text{Cash} + \text{Loans} + \text{Collateral} = \text{Deposits} + \text{Wholesale} + \text{Equity}. \tag{1.7}$$

Balance-sheet symbols: Cash is unrestricted cash; Loans is recognised loan principal; Collateral is the restricted asset from posted collateral; Deposits and Wholesale are customer-deposit and drawn wholesale-funding liabilities; Equity is the residual equity stock. All refer to the same date and are in KES millions in this table. NII means net interest income over the period.

Equity changes by realised profit less distributions. Posting collateral uses cash and creates a restricted asset; it is not treated as an expense. Charge-off reduces loans and profit; recoveries relate to already charged-off balances. Facility limits can cause a case to fail rather than creating unrestricted funding.

The Decimal residual tolerance is 10^{-20} in the chosen unit. A regression rescales the same path from millions to KES and checks that continuation and economic results are preserved. The scenario excludes impairment allowances, tax, fair-value accounting, capital ratios and loss of interest before the chosen charge-off date. It therefore demonstrates a cash/equity bridge, not IFRS financial statements or an LCR/NSFR return.

Default timing and discounted recovery are separate calculations

Conditional default hazards 10% then 20% produce marginal default probabilities **10% and 18%**, with 72% survival. Summing conditional hazards would be wrong.

A separate fixed-rate example contracts to collect 1,060 after one year at original EIR 6%. With 90% probability it collects as promised; with 10% probability it receives only 300 after two years. The expected discounted shortfall is:

$$0.1\left(1000-\frac{300}{1.06^2}\right)=73.3001068. \tag{1.8}$$

In this formula, 0.1 is the probability of the default path, 1,000 is initial carrying value, 300 is the recovery at year two, 1.06 is one plus the original annual effective interest rate (EIR), and exponent 2 is the elapsed two years. Amounts are separate currency units, not the millions in the bank table.

This illustrates recovery timing. It does not choose stage, define default or infer cure from cumulative-minus-current delinquency. Those credit states require the existing Underwrite/Mwendo definitions and event histories.

A constrained hedge choice

The toy exposure is (800,1200) currency/bp on two axes. Candidate per-unit sensitivity vectors are (100,10) and (10,200); positions are integers from zero to ten. Costs are 100/200 currency per unit and collateral needs 500/1,500. Exhaustive search over 62 feasible choices under a 10,000 collateral-cash budget selects five units of each, leaving sensitivity (250,150), cost 1,500 and collateral 10,000.

PV01 denotes value sensitivity to a one-basis-point movement on the selected axis, in currency per bp. The first normalising 1,000 below is in currency per bp; the cost normaliser is 1,000 currency; the penalty weight 0.001 is dimensionless. The sum is over both sensitivity axes.

The objective is dimensionless: $\sum_j(\text{residual PV01}_j/1000)^2+0.001\,\text{cost}/1000$. The scale and penalty are explicit policy assumptions. This is a transparent limited optimisation problem, not an optimal market hedge, margin model, option valuation or capital-relief calculation.

The collateral constraint excludes transaction costs. For the selected hedge, collateral plus transaction cost requires 11,500 currency if both are paid at inception; the solved objective is 0.0865.

Implementation: scenarios.py.

6. STP demonstrated locally

STP means straight-through processing. SQLite is the local database; ID means identifier, and API means application programming interface. Processing amounts use integer minor units: 100 minor units = KES 1; rates in the processing payload use integer basis points.

The SQLite case follows a synthetic 250,000 KES deposit placement through proposal, approval, order, confirmation, booking, failed settlement, retry and reconciliation. Eight accepted events produce four journal lines. Reopening the database and replaying settlement adds neither a second event nor another posting.

Role violations, changed confirmed economics, duplicate IDs with different payloads, wrong settlement amounts and wrong retry references fail. Journal reconstruction checks accounts and amounts against accepted events. A deliberately doubled but still balanced ledger fails that reconstruction: a zero trial balance alone is insufficient.

Actor labels are not authenticated identities. No vendor, broker, bank or supervisor is connected. The original demonstration's scope is in operations.py and the operations methodology. The lifecycle extension adds

partial fills/settlement, cancellation/amendment and approved accrual corrections. Distributed recovery and production security still require operational implementation.

7. Verification boundary and next work

Tests check independent identities, historical index agreement, analytic curve sensitivities, unit scaling, probability mass, money conservation, known parameter recovery, future-data separation and durable workflow behaviour. They do not prove data quality, causal treatment effects, legal enforceability, accounting approval or production performance.

The 18-item reconciliation now distinguishes the implemented OAS, liquidity, derivative-exposure/capital components, hedge-accounting and connected-cash fixtures from broader unfinished research and institutional work. Live interfaces and empirical calibration remain open. The cross-project report identifies current owning-project implementations and their remaining requirements.

8. Expanded reference cases — 1 October 2026

The original cases above retain their numerical baseline. These additions turn further review requirements into executable examples:

Additional worked case	What to read
Dated balances, caps/floors, fixing-correction approval and partial settlement	Lifecycle cases
Curve-fitted rate paths, prepayment/OAS, priced hedges, collateral funding and Merton calibration	Options and collateral
Classified LCR/NSFR, SA-CCR, CVA/CS01, reduced BA-CVA and FRTB GIRR delta	Regulatory cases
Fair-value/cash-flow hedge journals and borrower-to-owner cash interfaces	Accounting and connected finance
What is implemented and what still needs development or external evidence	Updated implementation register
Product candidates and reusable acceptance fixtures	SokoIntel map

These methods complement CBK's benchmark-plus-premium pricing framework. Each module declares its own financial quantity, instrument and intended use; a capital component or cash illustration is labelled with that scope.

Part II. Funding, deposits and processing

28 September 2026 | Synthetic reference implementation for R02, R03 and R09

This laboratory turns three useful ideas from the supplied material into inspectable computations: internal funding attribution over a product's life, estimation of deposit repricing and runoff from a longitudinal panel, and a replay-safe transaction workflow. It supplies local examples and tests. It does not establish empirical Kenyan behaviour, executable market terms, regulatory compliance, accounting-policy approval or vendor connectivity.

1. Dated product-level FTP

Definitions

For each half-open interval `[start, end)`, use actual calendar days divided by 365. Partition at every balance or reset event. Inputs carry both the last rate-reset date and the liquidity/commitment horizon. A reset changes the rate; it does not necessarily release the bank from its remaining funding commitment.

Let $H_D = \text{drawn} \times \text{days}/365$ and $H_U = (\text{limit} - \text{drawn}) \times \text{days}/365$.

Quantity	Calculation
Customer income	$H_D r_{\text{customer}} + H_U f_{\text{customer}}$
Drawn internal FTP	$H_D r_{\text{FTP}}$
Internal contingent-liquidity allocation	$H_U r_{\text{liquidity}}$
External funding expense	$H_D r_{\text{external}} + H_U f_{\text{external}}$
Business-unit margin	Customer income less both internal charges
Treasury margin	Both internal charges less both external expenses
Consolidated margin	Customer income less external funding expense

The rate symbols r_{customer}, r_{FTP}, $r_{\text{liquidity}}$ and r_{external} denote the matching annual decimal rates in the table; f_{customer} and f_{external} denote the annual commitment-fee and commitment-cost rates. Adding business-unit and Treasury margins eliminates the internal transfers exactly before presentation rounding. Changing an internal FTP rate alone changes departmental attribution without changing consolidated margin. A conditional-default credit conversion factor is not used as the interest-bearing balance or as a universal liquidity-cost weight. Undrawn exposure-time and external contingent-funding cost are stated directly.

The calculation uses `Decimal` for monetary accumulation and rounds only the reported amounts to two decimal places. Displayed interval subtotals can differ by a cent from an independently rounded total. No discounting, credit losses, capital hurdle, tax or operating expense is included. The result is a funding contribution, not full economic profit or an APR. Principal repayments/draws change interval balances; principal is not counted as income.

Worked year

The examples cover 1 January 2027 to 1 January 2028, with quarter lengths 90, 91, 92 and 92 days. All values are synthetic currency units, interpretable as KES for teaching.

Product	Customer income	Internal FTP including commitment	Business margin	Treasury margin	Consolidated margin
Fixed 1,000,000 bullet	150,000.00	90,000.00	60,000.00	10,000.00	70,000.00
Amortising 1m / 750k / 500k / 250k	93,390.41	56,034.25	37,356.16	6,226.03	43,582.19
Quarterly floating 1,000,000	148,794.52	93,794.52	55,000.00	5,000.00	60,000.00
Revolver, actual synthetic path	104,054.79	70,104.11	33,950.68	7,428.77	41,379.45
Revolver, frozen ex-ante path	88,000.00	59,000.00	29,000.00	8,000.00	37,000.00

The revolver has a 1m limit and quarter-opening drawings of 400k, 800k, 600k and 900k. Its ex-ante case holds drawings at 600k and initial rates constant. It earns a 1% annual fee on undrawn commitment, allocates 2% internally for contingent liquidity, and incurs a separately assumed 0.75% external commitment cost. The actual-path contribution exceeds plan by 4,379.45, while Treasury's allocated margin falls by 571.23. Those are changes in the explicitly stated paths, not proof of a better pricing policy.

R02 advancement: dated fixed/amortising/floating/revolving cases, separate reset and liquidity horizons, contingent funding costs, internal elimination and actual-versus-plan attribution now execute. Remaining: optional draw/prepayment valuation, calibrated funding curves, discounted lifetime economics, transaction accounting and evidenced product conventions.

2. Deposit repricing and balance evidence

Synthetic panel and timing

`synthetic_deposit_panel()` provides two segment-level monthly panels, with opening observations in January 2023 and monthly changes through January 2026. The two names are labels of invented data-generating processes. They are not observed segment statistics or estimates from Kenyan accounts.

Training uses February 2023 through December 2024: 23 monthly changes per segment. Future holdout uses January 2025 through January 2026: 13 changes per segment. The January 2023 opening row is excluded from estimation. A known change in the synthetic immediate upward response begins in January 2025. The model is frozen at the training cutoff and is not refitted on this future segment.

The prespecified one-month distributed-lag equation, in basis points, is:

$$\begin{aligned}\Delta r_t = \alpha &+ b_{\text{up}}\,\max(\Delta m_t, 0) + b_{\text{down}}\,\min(\Delta m_t, 0) \\ &+ \ell_{\text{up}}\,\max(\Delta m_{t-1}, 0) + \ell_{\text{down}}\,\min(\Delta m_{t-1}, 0) + \varepsilon_t.\end{aligned} \tag{2.1}$$

Here Δr_t is change_in_paid_rate, α is intercept, Δm_t is market_change, ℓ_{up} and ℓ_{down} are lag_{up} and lag_down, and ε_t is error; the index t denotes a monthly observation..

Closed-form OLS uses $(X^{\mathsf{T}}X)^{-1}X^{\mathsf{T}}y$, with pivoted elimination and singular-design rejection. A one-month lag is a modelling assumption here; the exercise estimates its coefficient, not the optimal lag length. Fit

and validation are segment-specific. The reported standard errors use the simple independent, constant-variance residual formula. They are diagnostic only: real panels require autocorrelation/heteroskedasticity treatment, account/segment dependence, uncertainty assessment and a justified time-series specification.

Segment	Immediate up beta	One-month up lag	Combined up response	Combined down response
Operating, synthetic	0.15027	0.10202	0.25229	0.14948
Rate-sensitive, synthetic	0.50027	0.20202	0.70229	0.54948

Training RMSE is approximately 0.121 basis points for each segment. It rises to 4.209 basis points in the deliberately changed holdout regime. This is useful evidence that a good in-sample fit need not survive a change in the repricing process. It does not establish a forecasting result on bank data.

Holdout predictions condition on the market-rate changes observed in those future months. They test the frozen repricing relationship out of time; they are not an ex-ante forecast of the market-rate path itself.

Balance decay and migration

Each original tagged cohort starts with 1m. Every monthly record separately stores withdrawals leaving the bank, migration to another product at the same bank, new inflows and closing balances.

$$C_{t+1} = C_t - W_t - M_t.$$

$$B_{t+1} = B_t + I_t - W_t - M_t.$$

In the two balance identities, C_t is the original tagged-cohort balance, B_t the original-product balance, W_t withdrawals, M_t same-bank migration and I_t new inflow; all are money for month t. The original cohort cannot be replenished by new sales or inflows. Same-bank migration reduces the original product's available funding but is separately visible in the bank-retained cohort. This simple generator keeps migrated funds at the same bank and does not model their subsequent withdrawal or interest expense. It is therefore not a complete bank deposit model.

A constrained ladder using the observed retention path

The panel's original-product retention fractions at 1, 3, 6 and 12 months are 0.9810, 0.94415, 0.89162 and 0.79573. The ladder uses these *observed synthetic fractions*, not the estimated rate beta, as capacity bounds.

The solver enumerates 5% weight increments at overnight, 3m, 6m and 12m maturity. It maximises the weighted illustrative yield subject to:

- Weights are nonnegative and sum to one.
- At least 20% is available overnight.
- Weighted average maturity is at most six months.
- At each observed horizon, the fraction still locked beyond that horizon does not exceed observed retention multiplied by a stated policy haircut.

With synthetic yields of 4%, 5%, 5.5% and 6%, a 10% retention haircut gives 20% overnight, 60% at six months and 20% at twelve months: 5.3% annualised yield. A tighter 25% haircut requires at least 26.425% overnight from the one-month retention bound; the 5% grid makes that 30%. The resulting optimum is

30% overnight, 40% at six months and 30% at twelve months, yielding 5.2%. The policy assumption and the retention estimate are both visible and separately changeable.

These are terminal maturity rungs, not rolling multi-year bond buckets. The example does not extrapolate a five-year maturity from three years of synthetic data or infer duration from beta. It is an illustrative constrained cash ladder rather than a calibrated replicating portfolio minimising bank-wide NII/EVE error. Regulatory caps, collateral, credit risk, liquidity eligibility, transaction costs, reinvestment and actual execution remain separate work.

R03 advancement: segmented asymmetric lag estimation, explicit cohort flows, future validation and a data-constrained ladder now execute on synthetic data. Remaining: empirical longitudinal data, selection/attrition handling, robust uncertainty, competing products, multi-cohort validation, lag/regime selection and validated longer-horizon retention forecasts.

3. Durable local STP demonstration

`STPWorkflow(db_path)` models one fixed-deposit placement. It does not represent a swap execution facility or a vendor API. Amounts are integer minor units, and rate is an integer number of basis points. The demonstration's 25m minor units represent a 250,000 KES placement.

The accepted sequence is:

`PROPOSE → APPROVE → ORDER → CONFIRM → BOOK → SETTLE → RECONCILE`.

Approval may instead end at `REJECTED`. A booked placement may record `SETTLEMENT_FAILED`, with no cash journal; a subsequent settlement requires a new event ID and a reference to that failure. The worked case includes this failed-and-retried route.

Control	Executable behaviour
Durable identity	SQLite event IDs; proposal/order/trade/confirmation/settlement IDs persist between process restarts
Frozen economics	Confirmation must equal approved currency, amount, dates, rate, instrument, counterparty and evidence versions, and match the order ID
Role separation	Proposer, approver, dealer, backoffice and reconciler are distinct mapped roles; a proposer cannot approve their own proposal even if their configured role later changes
Idempotency	Replaying the same event, actor, kind and payload returns the stored result; reusing its ID with changed content fails
Transaction boundary	Event, state projection and journal commit in one SQLite transaction; failed validation rolls back
Settlement	Cash amount must equal the approved full placement; a failed attempt produces no cash posting
Reconciliation	External cash and position totals must match the placement; the journal must match independent reconstruction from accepted events

The booking entries are debit placement asset and credit settlement payable. Settlement debits settlement payable and credits cash. For the 250,000 KES example, the incremental minor-unit ledger records `placement_asset +25,000,000`, `settlement_payable 0`, and `cash −25,000,000`. These are changes attributable to this placement, not the bank's opening or closing cash balance. The external `bank_cash_cents` field is its signed statement movement for this transaction. The four journal lines sum to zero, and their accounts and

amounts must also match the accepted event history. A test deliberately doubles every journal amount: the trial balance remains zero but the event reconstruction correctly rejects it.

Eight accepted events, including the failed settlement, produce four journal lines. Reopening the SQLite file and replaying the successful settlement leaves these counts unchanged. Invalid role, self-approval, rejected proposal, changed terms, wrong amount and wrong retry-reference cases are tested.

Actor names are supplied by the local application and are **not authenticated identities**. Database access is not tamper-proof. Rejected API attempts are rolled back rather than retained in a separate security audit log. The simulation has no external acknowledgement polling, partial fills, cancellation/amendment lifecycle, real settlement bridge, collateral engine or accrued-interest postings. Those features require their own interface, authority and accounting specifications.

R09 advancement: a durable, tested local reference chain now demonstrates identifiers, approval, confirmation, booking, failed settlement, restart, duplicate protection and reconciliation. Live interfaces, actual product eligibility and institutional acceptance remain unverified.

4. Verification and integration

Install `operations.py` in `research_lab/` and `test_operations.py` in `tests/`. The module depends only on the Python standard library. `operations_examples()` returns JSON-ready results for a shared build; it uses a temporary SQLite database for its demonstration. Direct use of `STPWorkflow` accepts an explicit durable database path.

From the repository root, run:

```
python -m unittest discover -s tests -p test_operations.py -v
```

Tests independently check a full-year funding identity, amortising exposure-days, changing-rate expenses, leap-year days, pure transfer-price invariance, known OLS coefficient recovery, future-data separation, balance identities, constrained ladder bounds, restart safety and independently reconstructed accounting entries. These checks establish the stated calculations and control behaviours; they do not make the source financial material fully validated or complete every item in the remaining-work register.

Part III. Symbols and units

1 October 2026. Reference for the worked examples and the accompanying operations methodology. A symbol's definition applies within its named section. Calculations retain full precision and round at the stated reporting step.

1. Common conventions

Notation	Meaning and units
KES	Kenya shillings. A table explicitly marked KES millions uses one unit for KES 1,000,000.
Annual decimal rate	A quoted 10% enters arithmetic as 0.10. Interest conventions determine how that annual quote accrues.
bp / bps	Basis point(s): 1 bp = 0.01 percentage point = 0.0001 in decimal-rate units.
ACT/365	Actual elapsed calendar days divided by a fixed denominator of 365. Leap-year intervals still use 365 in this particular convention.
`[start, end)`	A half-open interval includes the start date and excludes the end date.
Currency-years	Money multiplied by elapsed years. KES 1,000,000 outstanding for half a year contributes KES 500,000-years.
`i`, `j`, `n`	Integer indices. Their exact roles are defined below; they are not interest rates by themselves.
Sum / `sum`	Add the indexed terms. The large sigma in a displayed formula denotes this operation.
Product	Multiply the indexed terms. The large pi in the compounding formula denotes this operation.
$\sqrt{x}$	Nonnegative square root of x.
$\ln(x)$ / $\log(x)$	Natural logarithm; the inverse of the exponential function.
$\exp(x)$	Exponential function with base e.
$\max(x, 0)$ / $\min(x, 0)$	Positive part of x / nonpositive part of x. The latter retains the negative sign.
x^2, x^{-1}	Square / reciprocal of a scalar; for a matrix the exponent -1 means its matrix inverse.
10^{-8}, 10^{-20}	Powers of ten: numerical tolerances, not percentages. The accompanying text supplies the unit.
`m`, `k` in amount labels	Million and thousand, for example 1m = 1,000,000 and 400k = 400,000. These suffixes are distinct from the pricing variable k.

2. Benchmark compounding

Symbol	Meaning	Unit / convention
P	Principal held constant during the illustrated coupon period	KES
i	Index of a mapped benchmark observation interval	Integer
r_i	Annual decimal benchmark fixing assigned to interval i	Annual rate
n_i	Number of calendar days for which that observation contributes to the compounded factor	Days

Symbol	Meaning	Unit / convention
B	Day-count denominator	365 days per convention-year in the examples
d	Actual number of days in the contractual interest period	Days
G	Gross accumulation factor, product of all terms 1 + r_i n_i/B	Dimensionless; includes principal factor 1
A	Benchmark accrual factor G - 1	Dimensionless; interest factor only
$I_{\text{benchmark}}$	Benchmark interest P times A	KES
k	Annual simple contractual margin above the benchmark	Annual decimal rate
I_{margin}	Simple margin interest P k d/B	KES
F	A symbol used inconsistently in other manuscripts; deliberately not used as a calculation variable here	Use the explicit fields gross_accumulation_factor and benchmark_accrual_factor

The observation weights n_i need not sum to contractual d when a profile shifts observation intervals. The benchmark uses its mapped weights; simple margin uses contractual d. The lookback example must therefore retain both calendars. For the historical weekend, n_1 = d = 3 and B = 365.

The annualised period benchmark rate is A times B/d. The saved JSON field `factor` contains A, the accrual factor, rather than G. It should therefore be read with the field definition instead of inferred from its short name.

The published index is a dimensionless accumulated index level. The benchmark factor inferred from two levels is end index / start index - 1. On principal P, multiply that factor by P. Index publication rounding can create a small difference from fixing-based arithmetic.

In the historical example, the multiplication is P = 1,000,000; r_1 = 0.094618; n_1 = 3; B = 365; k = 0.03; d = 3. Benchmark and margin interest sum to KES 1,024.26 after rounding once to two decimals.

3. Curves, forward rates and value sensitivity

Symbol	Meaning	Unit / convention
D_n / D_j	Present value of one currency unit payable at annual node n / j	Dimensionless discount factor
$D(t)$	Discount factor at time t, including between nodes	Dimensionless; D(0) = 1
q_1	One-year simple deposit quote used to obtain D_1	Annual decimal rate; a one-year accrual
c_n	Annual par coupon quote for the unit-price instrument maturing at year n	Annual decimal rate; annual coupon intervals
n	Final annual coupon number / maturity year in the bootstrap	Integer
j	Coupon or payment index in the applicable sum	Integer
t_j	Time of payment j measured from valuation time	Years; t_0 = 0
T	Final payment time of the floating note	Years; equals its last t_j
$D_P(t)$	Factor from the projection curve used to infer forward coupons	Dimensionless; subscript P means projection, not principal
$D_D(t)$	Factor from the discount curve used to value payments	Dimensionless; subscript D means discounting

Symbol	Meaning	Unit / convention
f_j	Simple forward rate for interval from t_(j-1) to t_j	Annual decimal rate
N	Floating-note principal or notional repaid at T	Currency
PV	Present value of the stated cash flows	Currency
DV01 / PV01	Change in present value associated with a one-basis-point movement in the specified rate or curve node	Currency per bp

The bootstrap assumes a price of one, annual coupons and unit year fractions. Thus c_n times the sum of earlier discount factors is the present value of earlier coupons, and the final payment is 1 + c_n. A different coupon calendar needs its own accrual fractions.

The curve table's repricing error is calculated price minus one, a dimensionless unit-price residual. It differs from the fee-yield solver's present-value residual, which is measured in currency.

The reported key-rate DV01 is positive when raising that node's continuously compounded zero rate reduces value. It uses central finite differences: value after a downward bump less value after an upward bump, divided by twice the decimal bump size, then multiplied by 0.0001. The bump is initially 0.0001. The hedge example uses sensitivities to two specified axes; it does not assume those axes identify traded instruments.

4. Pricing, capital and yield

Symbol	Meaning	Unit / convention
H	Expected collectible interest-bearing balance integrated over the pricing horizon	Currency-years
k	Annual simple margin solved from the pricing identity	Annual decimal rate
C_{fund}	External funding expense over that same horizon	Currency
C_{op}	Allocated operating cost over that horizon	Currency
$EL_{\text{principal}}$	Expected loss of principal over that horizon	Currency; excludes missing coupon already removed from income
C_{cap}	Required return on allocated economic capital over that horizon	Currency; supplied directly to the calculation
Fee	Fee income included in the pricing identity	Currency
$I_{\text{benchmark}}$	Expected collectible benchmark income used by the pricing identity	Currency; calculate from the actual eligible balance path
EC	Allocated economic capital	Currency stock
h	Annual required return on economic capital	Annual decimal rate
τ	Length of the pricing horizon	Years
h EC	Capital return in the original one-year example	Currency for that one-year horizon
h EC τ	Simple capital return when EC and h are constant over a general horizon	Currency; dated capital changes need a dated sum
y	Annual effective cash-flow yield in the separate fee example	Annual decimal yield; distinct from regression vector y below

For a one-year constant-capital example, C_{cap} = h EC = 0.20 times 50,000 = KES 10,000. For other horizons

the program still expects C_{cap} in money. It does not infer a time horizon from an annual hurdle. With changing capital, accrue the required return on each dated capital balance under the elected convention.

RAROC is risk-adjusted return on capital. Here it is the one-year pre-tax return after funding, operating cost and expected principal loss, divided by EC. The numerator is KES 10,000 and EC is KES 50,000, giving 20%. Economic surplus subtracts C_{cap} from that numerator and is zero.

The fee-yield equation discounts KES 1,140,000 received after exactly one year to KES 990,000 net advanced: 990,000 = 1,140,000/(1+y). The one-year exponent is 1. The resulting yield is 15.151515%; statutory APR requires its own defined costs and legal convention.

5. Dated funding transfer pricing

Symbol / field	Meaning	Unit / convention
H_D	Drawn balance times the interval's actual days / 365	Currency-years; D denotes drawn
H_U	Undrawn commitment, limit less drawn balance, times actual days / 365	Currency-years; U denotes undrawn
drawn	Funded principal outstanding in an interval	Currency
limit	Approved total facility ceiling in an interval	Currency
days	Calendar days in the half-open interval	Days
customer rate	Interest rate earned on drawn principal	Annual decimal rate
customer commitment fee	Fee rate earned on undrawn commitment	Annual decimal rate
drawn FTP rate	Internal funding rate charged on drawn principal	Annual decimal rate
internal liquidity rate	Internal charge on undrawn commitment	Annual decimal rate
external drawn rate	External cost of funding drawn principal	Annual decimal rate
external commitment cost rate	External cost assigned to maintaining undrawn funding capacity	Annual decimal rate
reset_date	Most recent rate-reset date known by the interval start	Calendar date
liquidity_horizon	Date through which the funding commitment must be supported	Calendar date

Multiply each rate by its matching H_D or H_U to obtain interval income or expense in money. Sum intervals before rounding. A reset date and a liquidity horizon have separate roles. The business-unit margin plus Treasury margin equals consolidated contribution because the internal transfers cancel.

EAD means exposure at default, a currency amount conditional on default. A CCF, or credit conversion factor, maps an undrawn amount to an assumed default-time draw. Neither quantity replaces the actually drawn principal on which this example earns interest.

6. Deposit regression and retention

The paid-rate equation uses changes measured in basis points. It is estimated separately for each synthetic segment.

Symbol / field	Meaning	Unit / convention
change_in_paid_rate	Current month's change in the rate paid on deposits	bp
market_change	Current month's change in the selected market rate	bp
previous_market_change	Previous month's market-rate change	bp; zero for the first change record in this fixture
intercept	Fitted change when all four market-change regressors equal zero	bp
b_{up} / b_{down}	Immediate response to positive / negative market changes	Dimensionless bp per bp
lag_{up} / lag_{down}	One-month delayed response to positive / negative market changes	Dimensionless bp per bp
error	Observed paid-rate change less the fitted change	bp
beta	A rate-response coefficient; also shorthand for the vector of fitted coefficients	Components have the units specified above
X	Design matrix: a ones column, current positive and negative market changes, and lagged positive and negative changes	23 rows by 5 columns for each training segment
X′	Transpose of X, exchanging rows and columns	5 rows by 23 columns
y in OLS	Column vector of observed paid-rate changes	23 rows by 1 column; bp
$(X^{\mathsf{T}}X)^{-1}X^{\mathsf{T}}y$	Ordinary least-squares coefficient estimate	Five coefficients; assumes X has full column rank
RMSE	Root mean squared error: square root of the average squared observed-minus-predicted change	bp; averaged across the evaluated rows

Combined up response is b_{up} + lag_up; combined down response is b_{down} + lag_down. A downward market change is negative, so a positive b_{down} implies a negative contribution to the paid-rate change. The combined responses describe this chosen two-term lag structure. They do not measure balance retention or deposit duration.

The coefficient covariance diagnostic uses the residual sum of squares divided by 23 - 5 = 18 training degrees of freedom, multiplied by $(X^{\mathsf{T}}X)^{-1}$. Reported standard errors are square roots of its diagonal entries. This diagnostic assumes independent, constant-variance residuals. RMSE uses the number of evaluated rows in its averaging denominator, not those degrees of freedom.

The retention bounds pool the two initially equal 1,000,000 cohorts. At each horizon, add the surviving original-product cohort balances and divide by their total initial 2,000,000. This retains the original tags and excludes new inflows.

Retention / ladder term	Meaning and units
original cohort	Deposits tagged at the starting date; opening and closing balances are currency amounts. New inflows do not replenish this tag.
original product	The initial deposit product, including subsequently added inflows; its balances are currency amounts.
withdrawals	Money leaving the bank, in currency.
migration	Money leaving the original product for another product at the same bank, in currency.

Retention / ladder term	Meaning and units
retention fraction	Original tagged amount still in the original product divided by its initial amount; dimensionless.
retention haircut	Policy reduction of the available retention bound; 10% means multiply the bound by 0.90.
ladder weight	Fraction of the total allocation assigned to one maturity; nonnegative and summing to one.
3m / 6m / 12m	Three-, six- and twelve-month terminal maturities. The suffix m here denotes months, not millions.
overnight rung	Allocation available without a term lock; represented as zero months in the ladder's maturity constraint.
weighted average maturity	Sum of weight times maturity in months; capped at six months here.
weighted yield	Sum of weight times the annual quoted yield; a simple annualised comparison.
locked fraction	Total weight whose maturity is strictly beyond the horizon being tested.

7. Bank balances, default and recovery

Symbol / quantity	Meaning	Unit / convention
Cash	Unrestricted bank cash at the stated balance-sheet date	KES millions in the table
Loans	Outstanding loan principal recognised in this simplified ledger	KES millions
Collateral	Posted collateral recorded as a restricted asset	KES millions; separate from available Cash
Deposits	Customer deposit liability	KES millions
Wholesale	Drawn wholesale funding liability	KES millions
Equity	Residual accounting equity in this simplified scenario	KES millions
NII	Net interest income: loan/cash interest less deposit/wholesale interest	KES millions over the period
hazard	Probability of default in a period conditional on survival to its start	Probability between 0 and 1
marginal default probability	Probability, from the starting date, of default in one particular period	Prior survival probability times that period's hazard
survival	Probability of reaching the next period without default	Dimensionless probability
EIR	Original effective interest rate for the separate fixed-rate shortfall example	6% annual effective rate

Cash + Loans + Collateral equals Deposits + Wholesale + Equity at every shown date. Cash interest and funding interest accrue on opening balances; end-period movements affect interest in the next period. This is the elected ordering of the synthetic case.

For hazards of 0.10 and 0.20, second-period marginal default is (1 - 0.10) times 0.20 = 0.18, and final survival is 0.90 times 0.80 = 0.72.

In the shortfall formula, 0.10 is the probability of the default path; 1,000 is the original carrying amount; 300 is the default-path recovery at year two; 1.06 is one plus EIR; and exponent 2 is two years. The other path pays the contractual 1,060 at year one and therefore contributes zero discounted shortfall. Amounts in this separate example are currency units, not the bank table's millions.

8. Hedge objective and transaction units

Term	Meaning	Unit / convention
(800, 1200)	Unhedged sensitivities along the two specified axes	Currency per bp
(100, 10), (10, 200)	Sensitivity reductions from one unit of each candidate hedge	Currency per bp per instrument unit
position	Integer number of units of a hedge; each ranges from 0 to 10	Instrument units
residual PV01	Original sensitivity less both selected hedge contributions, separately by axis	Currency per bp
cost	Total transaction cost of selected units: 100 per first hedge plus 200 per second hedge	Currency
collateral_cash	500 per first hedge unit plus 1,500 per second hedge unit	Currency
10,000 budget	Maximum permitted collateral_cash	Currency; transaction cost is outside this constraint
first 1,000 in objective	Sensitivity scale used to normalise residual PV01	Currency per bp
second 1,000 in objective	Money scale used to normalise transaction cost	Currency
0.001	Relative penalty weight on normalised transaction cost	Dimensionless policy choice
objective	Sum of squared normalised sensitivities plus weighted normalised cost	Dimensionless

Five units of each hedge use collateral of 10,000 and incur cost of 1,500. Residual sensitivities are (250, 150). The objective equals 0.25 squared + 0.15 squared + 0.001 times 1.5 = 0.0865. If collateral and costs are both paid initially, total cash needed is 11,500. The enumerated collateral constraint alone does not certify that total cash is available.

Processing notation	Meaning
STPWorkflow(db_path)	Class implementing the local straight-through-processing workflow; db_path names its SQLite database file.
event ID	Persistent unique event identifier. Replaying the same content with the same ID must not duplicate effects.
minor units / cents	One hundredth of KES in this fixture: 25,000,000 minor units = KES 250,000.
placement_asset +25,000,000	Increase in the placement asset, recorded using a debit-positive signed ledger.
settlement_payable 0	Booking credit and settlement debit have offset after successful payment.
cash -25,000,000	Credit to cash for this transaction; an incremental movement, not total bank cash.
bank_cash_cents	Signed external statement movement for this transaction in minor units.
rate in processing payload	Integer basis points, unlike the annual decimal inputs to the pricing formulas.
notional_cents / cash_cents / position_cents	Placement principal, settlement cash amount and reconciled position, respectively, in integer minor units.
rate_bps	Annual rate stored as integer basis points: 500 denotes 5%.
PROPOSE / APPROVE / ORDER / CONFIRM	Propose the placement, authorise it, issue an order and verify confirmed economics.

Processing notation	Meaning
BOOK / SETTLE / RECONCILE	Recognise the placement/payable, record payment and match the accepted events to journals and external totals.
SETTLEMENT_FAILED / REJECTED	A failed payment attempt / a proposal refused by the approval step.

9. Abbreviation key

Abbreviation	Expansion
KESONIA	Kenya Shilling Overnight Interbank Average
CBK / CBR	Central Bank of Kenya / Central Bank Rate
RegTech	Regulatory technology
FTP / ALM	Funds transfer pricing / asset-liability management
EL / ECL	Expected loss / expected credit loss
EAD / CCF	Exposure at default / credit conversion factor
EC / RAROC	Economic capital / risk-adjusted return on capital
EIR / IRR / APR	Effective interest rate / internal rate of return / annual percentage rate
OLS / RMSE	Ordinary least squares / root mean squared error
PV / NPV	Present value / net present value
NII / EVE	Net interest income / economic value of equity
DV01 / PV01	Dollar value / present-value sensitivity for a one-basis-point rate movement; expressed in the model's currency
STP / SQL / SQLite	Straight-through processing / Structured Query Language / the embedded database used here
API / ID	Application programming interface / identifier
IFRS	International Financial Reporting Standards
LCR / NSFR	Liquidity Coverage Ratio / Net Stable Funding Ratio
OAS	Option-adjusted spread
SA-CCR	Standardised approach for counterparty credit risk
CVA	Credit valuation adjustment; its valuation and capital measures remain distinct
FRTB / IRRBB	Fundamental Review of the Trading Book / interest rate risk in the banking book
SPV	Special-purpose vehicle
R02 / R03 / R09	Existing work-register identifiers for lifecycle FTP, deposit behaviour and STP respectively

10. Sources and software

The numerical baseline is the repository's 28 September 2026 checked-example output. The 1 October edition adds notation, unit definitions and explanatory clarifications. Calculation modules and baseline inputs are unchanged.

- RegTech-KESONIA Treasury repository: calculation source, tests and publication builders.
- CBK revised risk-based credit pricing model, August 2025: section 4 and Appendix I provide benchmark-pricing context. A contract still needs its applicable conventions.
- CBK KESONIA observations and compounded index: primary-source series for the historical cross-check. Printed historical inputs are retained in the worked example; live tables can change their displayed window.

The public-source links were checked on 1 October 2026. The saved historical fixture remains dated evidence. Other portfolios, quotes, deposit panels, assumptions and transactions in this publication are synthetic.

11. Symbols in the expanded cases

This reference defines the original three-part laboratory. Each October 1 extension includes its own complete definitions beside equations and in a symbol/unit table:

- Lifecycle: dated balances, accrual profiles, correction amounts and execution states.
- Options and collateral: lattice state prices, CPR/SMM, OAS, swap risks, collateral and structural credit.
- Liquidity and capital: HQLA, ASF/RSF, RC/PFE/EAD, CVA and GIRR terms.
- Accounting and connected finance: designation, cumulative reserve, journal and cash-handoff fields.

Symbols are local to their declared case; typed result fields preserve the economic meaning when exchanging outputs.

Part IV. Contract and execution lifecycle

Edition: 1 October 2026. **Case:** KESONIA-LIFECYCLE-001. **Inputs:** synthetic. **Currency:** Kenya shillings, KES. All annual rates are decimals in the executable inputs.

This companion implements four accumulated review requirements: a dated principal-change calculation; a corrected fixing carried through an approved adjustment ledger; a recovery-evidence decision gate; and partial execution/settlement with durable event identity. The existing benchmark, funding and SQLite workflow examples provide the starting interfaces.

1. State the contract before changing the balance

The case elects a **benchmark-index account with a simple margin on funded principal**. Cash invested or withdrawn at an event date changes the principal and benchmark-account value by the same amount. Benchmark return remains in that account; the simple margin is recorded separately. The contract permits withdrawal of funded principal only. Settling accrued return or closing the account requires a further settlement event outside this profile.

This explicit hypothetical convention makes the balance-event mechanics reproducible. Applying it to a customer loan requires that loan's actual balance/accrual terms. The case does not establish a general right to capitalise unpaid customer interest. The existing constant-principal calculator remains available for its stated profile.

Elected terms

Term	Value or rule
Period	1 September 2025 included to 8 September 2025 excluded
Opening principal	KES 1,000,000
Principal events	Add KES 200,000 at the start of 3 September; withdraw KES 300,000 at the start of 6 September
Benchmark observations	Synthetic annual rates of 8%, 9%, 10%, 11% and 12% for 1–5 September; the Friday fixing spans the weekend
Benchmark floor/cap	Applied to each selected fixing: minimum 0%, maximum 20%; these limits are assumed contract terms
Day count	Actual calendar days divided by 365
Observation mapping	Zero-lookback illustration; the function also accepts an explicit nonnegative business-day lookback and calendar
Margin	3% initially, then 4% from 4 September under a scheduled reset authorised before that effective date
Contract timezone	UTC; each date-only effective event starts at 00:00 UTC
Reset permission	Authority identifier and matching contract version required by 00:00 UTC on the effective date; exact-midnight approval is permitted, later approval is rejected
Principal-event order	Effective at the start of a subinterval, before that subinterval earns return
Rounding	Decimal arithmetic at 40 significant digits; cash output rounded once to two decimal places, half up

Observation-shift, a compounded customer margin, interest-payment allocation and alternative floor placement are separate contractual profiles. The function rejects an unsupported margin or balance convention. The illustrative zero-lookback fixture does not resolve the final CBK framework's five-day operating map for a live product.

Mathematics and all symbols

Symbol	Meaning	Unit
a	Start date of one original fixing span	Calendar date
s, e	Start and end dates of an event-created subinterval within that span	Calendar dates; s included, e excluded
r	Selected annual decimal fixing after its elected floor/cap	Annual rate
b	Day-count denominator, 365 here	Days per year
$J(t)$	Benchmark index level at date t, relative to the start of the fixing span	Dimensionless
$g_{s,e}$	Growth factor over the subinterval	Dimensionless
P_s	Funded principal after start-of-subinterval principal events	KES
V_s	Benchmark-account value after those events, including previously retained benchmark return	KES
c_s	Signed principal addition at s; negative for a principal withdrawal	KES
k_s	Authorised simple annual margin applicable from s	Annual rate
$I^b_{s,e}$	Benchmark return earned in the subinterval	KES
$I^m_{s,e}$	Separately accrued simple margin in the subinterval	KES
V_T, P_0	Terminal benchmark-account value and original funded principal	KES
I^b	Total benchmark return for the case	KES

Within one original fixing span, the index is interpolated according to that span's simple day weighting:

$$\frac{J(t)}{J(a)} = 1 + r\frac{t-a}{b}, \qquad g_{s,e} = \frac{1 + r(e-a)/b}{1 + r(s-a)/b}. \tag{4.1}$$

The date differences count actual calendar days. Principal events first update both funded principal and benchmark-account value by c_s. Then:

$$I^b_{s,e} = V_s(g_{s,e} - 1), \qquad V_e = V_s g_{s,e}, \qquad I^m_{s,e} = P_s k_s \frac{e-s}{b}. \tag{4.2}$$

The benchmark return reconciles to:

$$I^b = V_T - P_0 - \sum_s c_s. \tag{4.3}$$

Using index ratios is important when an event splits a weekend. A Friday-to-Monday fixing weighted for three days remains the same underlying factor. Merely inserting a zero-amount Saturday event does not turn it into extra daily compounding. An independent test also values the mid-span principal withdrawal by the benchmark-index units sold on the withdrawal date.

Calculated original ledger

The table rounds each displayed increment for readability; the final total uses unrounded increments.

Accrual dates	Funded principal, KES	Applied fixing	Margin	Benchmark increment, KES	Margin increment, KES
1–2 September	1,000,000	8%	3%	219.18	82.19
2–3 September	1,000,000	9%	3%	246.63	82.19
3–4 September	1,200,000	10%	3%	328.89	98.63
4–5 September	1,200,000	11%	4%	361.88	131.51
5–6 September	1,200,000	12%	4%	394.90	131.51
6–8 September	900,000	12%	4%	592.61	197.26

The case closes with KES 900,000 funded principal, KES 2,144.092488402175 benchmark return, and KES 723.287671232877 simple margin. Total interest is **KES 2,867.38** after cash rounding. Closing benchmark-account value is KES 902,144.092488402175; the separate margin is not included in that value.

2. A fixing correction becomes an approved adjustment

At the original 8 September information cutoff, the Friday fixing is 12%. A corrected 13% version becomes available on 9 September. The contract, principal events, margin schedule and cash-rounding policy are unchanged.

Calculation/posting	Interest, KES	Result
Original cutoff	2,867.38	Original result and evidence retained
Revised cutoff	2,949.68	Corrected fixing version, later cutoff and new result hash
Approved adjustment	82.30	Revised rounded amount less the already posted rounded amount
Posted accrual after adjustment	2,949.68	Original posting plus one adjustment

`correction_case()` produces the proposal and two linked result hashes. `AccrualPostingBook` separately requires a maker/checker approval bound to the exact calculation JSON before recording a posting. Its durable ledger retains the original event and adjustment. The correction must identify the current predecessor and a later information cutoff. Replaying the same posting returns its original receipt; a new message identity cannot justify reposting the same calculation.

The cash adjustment uses the difference between the two **rounded posted amounts**. Independently rounding the change in unrounded values can differ by a cent. The output therefore retains both the unrounded change and the posting delta. A fixing-only correction also requires an unchanged contract-and-event hash, so an altered margin cannot be presented as a source correction.

This demonstrates a local accrual-control ledger. Institution-specific accounts, tax, recognition dates, customer notices and approved general-ledger mappings remain inputs to the operational implementation.

3. Recovery evidence has a verification clock

The evidence fixture deliberately holds borrower default probability and contract terms outside the gate. It tests whether new recovery evidence may enter an analytical review.

Information cutoff	Source situation	Last verified recovery fraction	Gate result
1 September, 12:00	Original rights confirmed and verified	60%	Confirmed; eligible for analytical refresh
2 September, 12:00	Source unavailable	60%	Request evidence; no new analytical assumption
3 September, 12:00	Changed evidence received, verification pending	60%	Review pending
4 September, 12:00	Changed rights verified	30%	Review verified change; analytical refresh allowed

Evidence timestamps use Africa/Nairobi's stated UTC+03:00 offset in this fixture. The date-only accrual contract uses UTC, and reset authority is compared as a full timezone-aware instant against 00:00 UTC on its effective date. An approval later on that date cannot authorise the earlier interval. Each evidence record preserves economic effective time, availability time, verification time, source-result version and review owner. The age limit is an elected five-day policy parameter. Source failure does not refresh the verification date or prove deterioration. A known but stale record cannot authorise an analytical refresh.

The gate returns `contract_change_authorised = false` and `borrower_pd_changed_by_this_gate = false`. The purpose is to route a verified recovery change for the appropriate review. The existing Underwrite decision case supplies the broader borrower-authority and agreed-action workflow; this RegTech fixture is its narrow recovery-assumption interface.

4. Partial execution and partial settlement are separate

The durable `PartialSettlementBook` extends an already approved order handoff. It stores monetary fields as **integer minor units**; 100 minor units equal KES 1 in this case. Each persisted book fixes one currency and instrument claim. The default case is KES synthetic deposit placements. Orders for another currency or instrument are rejected; restarting a book with a different scope is also rejected. Every aggregate output carries the book currency and instrument.

Event	Position, KES	Cash paid, KES	Unsettled payable, KES	Live unfilled order, KES
Approved order for KES 10,000; acknowledgement unknown	0	0	0	10,000
Confirmed partial fill of KES 6,000	6,000	0	6,000	4,000
First settlement of KES 2,000	6,000	2,000	4,000	4,000
Cancellation of the unfilled quantity confirmed	6,000	2,000	4,000	0
Final settlement of KES 4,000	6,000	6,000	0	0

Here a position is the booked synthetic placement asset, a payable is confirmed but unpaid settlement cash, and live unfilled quantity is an instruction that can still execute. Cancellation removes the latter. It does not erase the asset or settlement obligation created by an existing fill.

The invariant is **position + signed cash change - settlement payable = 0**. The first partial settlement therefore reconciles as KES 6,000 - 2,000 - 4,000 = 0. The code rejects cumulative settlement above the

confirmed fill and rejects a repeated settlement business ID even when it arrives under a different event ID. Restarting SQLite and retrying the final event creates no second cash posting.

Additional failure and amendment tests

- Unknown external acknowledgement never creates authority to send a second economic order. A repeated event returns the existing record. A confirmed fill can establish that execution happened despite the earlier missing acknowledgement.
- Every order/quantity amendment needs independent maker/checker identities and an approval tied to the exact terms. Reusing an approval identity for changed terms fails.
- The amendment fixture changes unfilled quantity only. It cannot reduce the total below confirmed fills or modify a cancelled order. A fill referencing pre-amendment terms is rejected.
- Increasing a fully filled order restores the partially filled state and exposes the new unfilled capacity; the confirmed position and payable remain unchanged until further fill/settlement events.
- Confirmed cancellation needs authority for the exact cancellation action and external cancellation evidence. Existing fills remain settleable.
- Money in floating units, booleans, nonpositive amounts, overfills, oversettlements and changed-content retries fail before durable economic effects are recorded.

The adapters are local mocks. Real authentication, external status queries, venue message mapping, matching tolerances, accrued interest on the placement, legal settlement finality and recovery operations require the chosen institution's interfaces.

5. Reproduction and review traceability

The calculations and SQLite examples use the Python standard library and the existing benchmark module:

```
python -m unittest discover -s tests -p test_lifecycle_cases.py -v

python -c "import json; from research_lab.lifecycle_cases import lifecycle_examples;
print(json.dumps(lifecycle_examples(), indent=2))"
```

Accumulated requirement	Implemented artifact	Verification
Part 2 principal changes, caps/floors and explicit margin/reset conventions	`accrue_account`, `AccountProfile`, dated events	Existing constant-principal reference; independent index-units withdrawal; split-span invariance; permission and interval rejection
Part 4 original/corrected calculation and adjustment replay	`correction_case`, `AccrualPostingBook`	Same contract/events, exact-result approval, rounded reconciliation, predecessor order and restart-safe duplicate handling
Cross-project XB04 evidence freshness and recovery readiness	`freshness_gate`	No future verification leakage; outage preserves last verified state; stale/changed/no-change paths
R09 and Case F partial execution/settlement, unknown acknowledgements and amendments	`PartialSettlementBook`	Cumulative cash/position/payable invariants; cancellation retains confirmed obligations; approval/terms identity; restart/replay

The 22 lifecycle tests close these elected reference fixtures, including timezone-aware reset-authority checks at the UTC day boundary. Institutional contract interpretation, production authentication, whole-system concurrency and live vendor/ledger acceptance retain their separate completion evidence.

Part V. Options, hedges and collateral

1 October 2026. This addition supplies executable reference cases for R04, R05 and R06, plus a compact R12 structural-credit baseline. It is designed to be read alongside Worked examples, Symbols and units, and the remaining-work register.

The cases use synthetic inputs and standard-library Python. All monetary amounts are Kenya shillings (KES); all model rates are annual decimals. A displayed 0.01 means 1 percentage point or 100 basis points. The choice of KES as the reporting unit does not establish that these securities, hedge tenors, collateral agreements or model parameters are available or approved for any Kenyan institution.

1. What is executable now

Register row	Executable evidence	Completion boundary
R04, prepayment and OAS	Monthly additive short-rate tree fitted to every synthetic discount factor; amortizing pool; rate-responsive prepayment; a separate optimal-exercise calculation; bracketed numerical OAS recovery	Market volatility calibration, permissioned borrower histories, physical-to-valuation-measure treatment and institutional validation remain open
R05, hedge selection	Actual synthetic swap valuations at two tenors, 25 candidate portfolios, basis and curve-shape shocks, finite-difference slope/curvature, transaction-fee and initial-cash limits	Tradable quotations, transaction sizes, bid/offer, liquidity, execution and approved risk limits remain open
R06, collateral and clearing	Netting, threshold, minimum transfer, haircuts, dated and intraday calls, settlement lag, IM increases, borrowing and two accounting identities	An institution's enforceable agreement, eligible collateral, custodian/CCP rules, disputes, real intraday feeds and legal conclusions remain open
R12, structural credit	Terminal-debt Merton valuation, debt/equity identity, limiting cases and recovery of synthetic asset value/volatility from equity observations	Compound options, coupon/default barriers, empirical estimation, and comparison with reduced-form models remain open

No row is treated as an implemented institutional model merely because its reference calculation runs.

2. Reproduce the results

Run from the RegTech repository root with Python 3.10 or newer:

```
python -m research_lab.option_cases

python -m unittest discover -s tests -p test_option_cases.py -v
```

The first command writes JSON to standard output. To preserve it with explicit UTF-8 encoding:

```
python -c "import json; from pathlib import Path; from research_lab.option_cases import
option_examples; p=Path('research_lab/results/option_cases.json');
p.parent.mkdir(parents=True, exist_ok=True); p.write_text(json.dumps(option_examples(),
indent=2, allow_nan=False), encoding='utf-8')"
```

The implementation is research_lab/option_cases.py; the tests are tests/test_option_cases.py.

`option_examples()` returns ordinary JSON-serializable values and performs no file writes. Importing the module runs no examples. Twenty-nine tests cover numerical recovery, independent path enumeration, conservation of principal, option bounds, hedge constraints and funding/accounting identities.

3. R04: fit a curve before valuing the prepayment option

3.1 Synthetic curve and short-rate lattice

Use 36 monthly steps. The year fraction per step is $\Delta t = 1/12$. At maturity $t_i = i/12$ years, the input discount factor is:

$$D_i = \exp[-(0.085 + 0.003t_i)t_i]. \tag{5.1}$$

Here i is the month index, D_i is the price today of one KES paid at t_i, and $D_0 = 1$. The continuously compounded zero rate rises from 8.5% at the origin to 9.4% at year three. These are generated values, not observed KESONIA quotations.

The one-factor additive recombining tree has risk-neutral branch probability one half. At layer i, node j records j upward moves among i moves:

$$r_{i,j} = a_i + (2j - i)\sigma\sqrt{\Delta t}. \tag{5.2}$$

$r_{i,j}$ is the annual continuously compounded short rate for the following month. a_i is the fitted deterministic level, and $\sigma = 0.015$ is the assumed absolute short-rate volatility: 150 basis points per square-root year. This is a **Ho–Lee-style lattice**, with no mean reversion; it is not a calibration of Hull–White or Black–Karasinski parameters.

Let $q_{i,j}$ be the time-zero price of one KES contingent on reaching node (i, j). Begin with $q_{0,0} = 1$. Fitting the next discount factor gives:

$$a_i = \frac{1}{\Delta t} \log \left(\frac{\sum_j q_{i,j} \exp[-(2j - i)\sigma\sqrt{\Delta t}\, \Delta t]}{D_{i+1}} \right). \tag{5.3}$$

Each branch carries half of $q_{i,j} \exp(-r_{i,j}\Delta t)$ into its child node. Consequently, summing next-layer state prices reproduces D_{i+1}. The largest absolute discount-factor repricing error in the case is **2.22×10^{-16}**. The tests also fit negative-rate curves: the additive model allows negative nodes and does not silently floor them.

Fitting today's curve does not estimate volatility. Volatility must be supplied here; market-option calibration would require a separate quotation set and objective. The original Ho and Lee paper supplies the model lineage; the lattice equations above state the exact implementation used here.

3.2 Scheduled principal and prepayment

The synthetic pool starts with KES 1,000,000, a nominal annual coupon of 14%, monthly payments and a 36-month remaining term. Borrowers are homogeneous surviving loans. Prepayment removes a fraction of otherwise identical loans from the pool; each remaining loan retains its ordinary amortization schedule. There is no default, recovery lag, servicing fee or heterogeneous burnout variable in this example.

At a node, let B be opening surviving principal in KES, c the nominal annual coupon, $u = c/12$ the monthly coupon rate, and n the number of remaining payments. Scheduled payment per KES opening balance is $u/[1-(1+u)^{-n}]$; for $u = 0$ it is $1/n$. The scheduled-principal fraction g is this payment fraction minus u, clipped to the interval from zero to one to remove final-payment floating-point noise.

The assumed annual conditional prepayment rate is:

$$\text{CPR}_{i,j} = \min[0.60,\ 0.04 + 6\max(c - r_{i,j} - 0.025, 0)]. \tag{5.4}$$

CPR is a fraction of surviving principal prepaid over a year. Here 0.04 is background prepayment, 6 is the response coefficient to the positive refinancing incentive, 0.025 is the assumed refinancing markup over the short rate, and 0.60 is the cap. The refinancing rule is a deliberately simple valuation assumption. It is not an estimated response to actual mortgage offers, nor does the short rate alone reconstruct a contractual refinancing term rate.

Convert annual CPR to single monthly mortality (SMM) as:

$$\text{SMM}_{i,j} = 1 - (1 - \text{CPR}_{i,j})^{1/12}. \tag{5.5}$$

Interest is uB. Scheduled principal is gB. Prepayment is $\text{SMM}_{i,j}(1-g)B$, applied after scheduled principal. Closing balance is $(1-g)(1-\text{SMM}_{i,j})B$. The CPR/SMM relationship agrees with the convention described in MathWorks' mortgage-pool documentation.

Cash is paid at the month's end using the rate and CPR known at its start. Two balance arrays are propagated separately: probability-weighted balances for expected cash flows, and discounted probability-weighted balances for valuation. This preserves dependence between rates, prior prepayment and surviving balances. The aggregation is exact for this linear homogeneous-pool rule. The tests compare a six-month tree with exhaustive enumeration of all 64 paths. A model with borrower-specific refinancing friction or path-dependent burnout needs additional state variables.

3.3 Numerical OAS solve

The option-adjusted spread s is a constant annual continuously compounded spread added to each short rate for discounting. A payment CF_m at month m is valued as:

$$P(s) = E_Q\left[\sum_{m=1}^{36} CF_m \exp\left(-\sum_{i=0}^{m-1}(r_i + s)\Delta t\right)\right]. \tag{5.6}$$

$P(s)$ is the model price in KES; E_Q is expectation under the stated valuation measure; CF_m includes interest, scheduled principal and prepayment; and r_i is the rate along the path. The prepayment rule remains fixed during the spread solve. A larger spread therefore discounts positive cash flows more heavily and produces a lower price. MathWorks' OAS reference gives the general price-to-spread use; the assumptions and solver here are explicit and independent of that software.

For a recovery test, generate a target price using a known spread of 175 basis points, then pass only the price to bracketed bisection over spreads from -5% to +20%. The solver rejects an unbracketed target and must reach an absolute price error below KES 0.0000001 within 100 iterations.

Quantity	Reproduced value
Synthetic target price at 175 bp	KES 1,037,135.316792
Recovered OAS	174.999999999955 bp
Price residual	KES 0.000000005472
Solver iterations	41
Price at 75 bp	KES 1,049,720.716375
Price at 275 bp	KES 1,024,771.515460
Total expected principal paid	KES 1,000,000
Final expected principal	KES 0, within floating-point tolerance

The monthly cash-flow rows in JSON expose interest, scheduled principal, prepayment, opening/closing balance and the principal reconciliation residual. A known-price recovery test establishes implementation consistency; it does not establish empirical model accuracy or the economic uniqueness of an OAS inferred under different behavioural assumptions.

3.4 Exercise and baseline checks

A separate backward-induction calculation treats the borrower as choosing frictionless whole-loan repayment at par at the start of months 2–36. It compares the lender's continuation value with outstanding principal and takes the smaller. Immediate time-zero repayment is excluded. Its purpose is to demonstrate an explicit exercise decision as distinct from the CPR response rule.

Model at zero OAS	Value, KES
Scheduled mortgage with no prepayment	1,072,762.685598
Stated rate-responsive CPR policy	1,059,308.030558
Frictionless optimal borrower exercise at the stated call dates	1,004,505.086704

The exercise engine identifies 469 exercise nodes, including nodes that may be unreachable after an earlier exercise; this is a count of the exercise region, not a count of borrowers or actual repayments. Its noncallable limit equals the scheduled cash-flow value from the fitted curve. With a zero coupon and positive deterministic rates, there is no optimal early exercise. These are useful economic checks independent of the OAS recovery calculation.

The CPR case assumes a valuation-measure prepayment process. A longitudinal physical-measure model of borrower decisions requires observed exposures, censoring/competing exits, out-of-time tests and an explicit treatment of unhedgeable behavioural risk before it can supply valuation assumptions. Curve fitting and mechanical spread recovery do not complete that empirical work.

4. R05: compare priced hedges and preserve basis risk

4.1 Instruments and price equations

The asset is a three-year annual-pay fixed-rate instrument: KES 1,000,000 principal and KES 110,000 coupon each year. Baseline continuous discount and projection zero rates are 8.5% and 9% respectively. Its base price is KES 1,053,996.793847.

Candidate hedges are annual pay-fixed/receive-floating synthetic swaps with two- and five-year tenors. The model uses annual unit accrual fractions, no principal exchange, deterministic projection from discount-

factor ratios, and a fixed rate set to give zero inception value. With payment year t, notional N, fixed rate K, projection factor $D_P(t)$, and discount factor $D_D(t)$:

$$V_{\text{swap}} = N \sum_{t=1}^{T} \left[\frac{D_P(t-1)}{D_P(t)} - 1 - K \right] D_D(t). \tag{5.7}$$

T is the integer tenor in years. V_{swap} is positive for a payer whose projected floating receipts exceed fixed payments after discounting. Both inception fixed rates are approximately 9.417428% in this flat projection example. Interest payment forecasts are synthetic annual-period amounts; an implementation for actual resets and compounding must use the instrument's fixing and accrual conventions.

Parallel shocks move both curves. The basis shock raises the projection curve alone by 100 bp. The twist adds $\ell(t-3)/3$ to both zero curves with $\ell = 0.01$. Here ℓ is an annual rate coefficient, and t is maturity in years. The asset responds only to its discount curve. These scenarios therefore preserve distinct projection, discount and maturity effects.

4.2 Finite feasible set and objective

Each tenor can have notional 0, 250,000, 500,000, 750,000 or 1,000,000, giving 25 portfolios. Two-year and five-year fees are respectively 0.10% and 0.15% of notional. Initial-margin assumptions are respectively 3% and 6%. The fee limit is KES 2,000; total initial margin plus fees must fit KES 70,000. Initial margin is restricted liquidity, while fees are an expense.

The chosen candidate minimizes the worst absolute combined asset-and-hedge value change **after fees** across four specified shocks. It is a finite-grid comparison, not a claim of continuous global optimality. It selects KES 750,000 of the two-year payer swap, no five-year swap, fee KES 750 and assumed IM KES 22,500. Total initial cash use is KES 23,250.

Stress	Asset change	Hedge change	Combined change after fee
Both curves +100 bp	-28,264.44	14,320.36	-14,694.09
Both curves -100 bp	29,085.87	-14,603.44	13,732.43
Projection curve alone +100 bp	0.00	14,533.51	13,783.51
Stated curve twist	1,296.58	-5,041.85	-4,495.27

All values are KES. The basis scenario explains why cancelling parallel duration alone is insufficient. The full JSON includes every feasible and infeasible candidate with its assumptions and scenario prices. The initial-cash constraint does not cap subsequent margin calls; those are addressed separately below.

4.3 Slope and convexity diagnostics

For a scalar parallel shock x, let $V(x)$ be repriced value and $b = 0.0001$ the 1 bp calculation bump. The signed slope is $[V(b) - V(-b)]/(2b)$; multiplying by 0.0001 gives KES change per bp. The curvature is $[V(b) - 2V(0) + V(-b)]/b^2$, measured in KES per squared decimal rate. The second-order contribution at a 100 bp shock is half this curvature times 0.01^2.

Diagnostic	Unhedged	Selected hedge plus asset
Signed parallel price change per bp, KES	-286.711459	-142.101793

Diagnostic	Unhedged	Selected hedge plus asset
Second derivative, KES per squared rate	8,213,664.78	5,382,893.58
Second-order contribution at 100 bp, KES	410.68	269.14

These derivatives are calculated from actual instrument prices. The test suite compares with an analytic zero-coupon benchmark and halves the bump to check convergence. Positive-duration bond terminology often quotes DV01 with the opposite sign; this module deliberately labels its measure `d_pv_per_bp` to keep the sign explicit.

5. R06: a value hedge can require more funding cash

5.1 Contract and time assumptions

This separate funding example uses a receive-fixed two-year swap on KES 250,000 and a pay-fixed five-year swap on KES 750,000, in one assumed legally valid netting set. It is a fixed portfolio chosen to illustrate offsetting contract marks; it is **not the candidate selected under the four-scenario objective above**.

The marked-value shocks occur at 09:00 and 15:00 between 5 and 8 October 2026. Rates fall cumulatively by 50, 100 and 200 bp. Prices use the preceding synthetic valuation equations with remaining tenors frozen across this short stress window. The dated ledger models call/settlement/funding timing; a complete dated security engine would also roll valuation dates, accrue coupons and use observed fixings. No coupon or maturity payment is included within this window.

Both cases begin with KES 40,000 cash. Funding costs 12% per year on ACT/365 including fractional days, charged on the previous event's borrowing over elapsed time. Costs are financed as they accrue. Required IM begins at KES 30,000 and rises according to the declared stress rule $30,000[1+|x|/0.02]$, where x is the parallel rate shift. This is a transparent synthetic IM scenario, not a CCP risk model. IM is posted immediately and remains a restricted asset.

Term	Bilateral illustration	Cleared illustration
Treatment of derivative variation	Collateral supporting an outstanding derivative mark	Final cash settlement to market; derivative mark reset to zero
VM exposure threshold	KES 5,000	Zero
VM minimum transfer amount	KES 500, applied to changes after pending transfers	Zero in this teaching ledger
VM settlement	Following weekday at 09:00	Same event, including intraday event
Collateral haircut	2% on the eligible collateral market value	2% on IM collateral; variation settles in cash
Reuse of restricted collateral	None	None

The positive bilateral threshold is an assumed legacy/otherwise-permitted contract term, not a claim that it is permitted for covered counterparties under current margin rules. The BCBS–IOSCO standard specifies zero VM thresholds for covered entities and separately addresses IM and transfer minima. Actual scope requires jurisdictional and counterparty classification.

The cleared example deliberately uses **final settlement variation**. Its accounting differs from collateralized-to-market clearing arrangements. CME's margining discussion explains the distinction between IM collateral

and settlement variation. An implementation must read the chosen product's legal terms rather than infer treatment from the word "cleared."

5.2 Calls, haircuts, netting and settlement

Let V be the net derivative value to the institution, θ the bilateral threshold, and h the collateral haircut. All three dollar-like quantities in this paragraph are KES; h is dimensionless. Required bilateral VM credit is $\max(-V-\theta, 0)$. Deliverable collateral market value equals required credit divided by $1-h$. The call is the change from posted collateral plus pending transfers. Changes below the minimum transfer amount remain uncalled until the threshold is met. IM is computed separately; it does not silently reduce VM exposure.

At the final 200 bp downward shock, individual swap marks are **+9,831.88** and **-67,136.37**, giving net value **-57,304.49**. Without the assumed netting agreement, the gross out-of-money mark is KES 67,136.37. The bilateral VM requirement is KES 52,304.49 after the KES 5,000 threshold; with a 2% haircut, posted VM collateral market value is KES 53,371.93. Pending transfers remain visible before the next morning's settlement. The code reports the residual exposure after settled VM and does not count a pending call as received collateral.

In the cleared case, cumulative final variation cash paid is KES 57,304.49. This payment realizes the derivative loss; it is not also booked as an owned collateral asset. IM market value is KES 61,224.49 against required credit of KES 60,000.

5.3 Funding and wealth reconcile

At every event, the implementation checks:

$$C_0 + F = C + R + S + J. \tag{5.8}$$

C_0 is initial cash; F is outstanding borrowing; C is available cash; R is restricted collateral market value; S is cumulative final variation settlement paid, zero in the bilateral case; and J is cumulative funding expense. Each is in KES. Borrowing is drawn only as needed to keep available cash nonnegative; available surplus first repays borrowing in the same identity.

The independent net-wealth check includes the asset's marked-value change and the derivative's remaining balance-sheet mark. In the bilateral case, the derivative liability remains outstanding alongside the collateral asset. In the cleared case, settled derivative variation has passed through cash, leaving a zero post-settlement derivative mark. Both routes reconcile to initial cash plus economic asset-and-hedge value change minus funding expense.

Final event, 8 October at 15:00	Bilateral	Cleared, final settlement
Asset value increase, KES	59,017.59	59,017.59
Derivative economic value change, KES	-57,304.49	-57,304.49
Combined economic value change, KES	1,713.10	1,713.10
Restricted collateral asset, KES	114,596.42	61,224.49
Cumulative final variation cash paid, KES	0.00	57,304.49
Cumulative funding expense, KES	23.80	40.91
Outstanding borrowing, KES	74,620.23	78,569.89

The hedge substantially reduces the marked-value movement while causing funding demand. This numerical case makes that distinction operational: asset valuation gains cannot automatically pay a margin call. It does not conclude that one clearing arrangement is universally cheaper; terms, offsets, eligible assets, timing and funding rates determine the result.

6. R12: structural-credit baseline and inverse calibration

The single-period Merton case assumes one terminal zero-coupon debt claim, no dividends, no bankruptcy cost and no default before maturity. Asset value is KES 1,500,000, face debt KES 1,000,000, asset volatility 30%, continuous risk-free rate 8.5% and maturity three years. The asset process and valuation assumptions are those of the standard structural baseline described in Merton's original paper.

For asset value A, debt face L, volatility σ_A, rate r, maturity T and standard normal distribution function Φ:

$$d_1 = \frac{\log(A/L) + (r + \sigma_A^2/2)T}{\sigma_A\sqrt{T}}, \qquad d_2 = d_1 - \sigma_A\sqrt{T}. \tag{5.9}$$

$$E = A\Phi(d_1) - Le^{-rT}\Phi(d_2), \qquad B = A - E. \tag{5.10}$$

E is equity value and B is risky debt value, both KES. The risk-neutral terminal default probability is $\Phi(-d_2)$. It is not an empirical one-year borrower PD. The implied equity volatility is $\sigma_E = A\Phi(d_1)\sigma_A/E$, measured per square-root year. The continuous credit spread over r is $-\log(B/L)/T - r$.

The example generates synthetic equity value and volatility from the known asset inputs, then recovers A and σ_A using nested bracketed solves. For each trial asset volatility, equity value determines an asset-value bracket between E and $E + Le^{-rT}$; the outer solve matches equity volatility within 10^{-12}. Recovered asset value is KES 1,500,000 and volatility 0.30 within the displayed solver tolerances. MathWorks' Merton documentation describes the corresponding equity/asset relation and inverse-estimation problem.

The tests separately verify $A = B + E$, the zero-volatility limit, terminal payoffs, and the increase in equity option value when asset volatility rises. `put_credit_discount` is $Le^{-rT} - B$, a present-value credit discount in KES. `equity_delta` is $\Phi(d_1)$, a dimensionless derivative. `credit_spread_continuous` is an annual decimal rate. At maturity or zero volatility, the implementation uses explicit limiting payoffs and the strict default event $A < L$ at maturity.

Actual balance-sheet assets are not automatically market asset value, and observed equity volatility is not asset volatility. Coupon debt, covenants, refinancing, early default, recovery costs, multiple maturities and compound options require a richer model. This new baseline advances R12's reproducibility; it does not close the broader research item.

7. Implementation and validation limits to carry forward

1. Supply permissioned borrower histories and a market option/curve quotation set before claiming empirical prepayment or volatility calibration.
2. Compare the CPR valuation assumption with the separate exercise model and test stability to refinancing costs, burnout and unhedgeable prepayment risk.
3. Replace frozen-tenor collateral stress marks with actual contractual event schedules, fixing accrual, date rolls and live valuation inputs.

4. Bind each derivative to its enforceable netting, collateral and settlement agreement; verify eligible assets, legal haircuts, segregation, reuse rights and call disputes.
5. Expand the hedge feasible set using executable quotations and stressed cash constraints, while preserving basis and convexity exposures.
6. Keep structural-credit valuation and empirical default forecasting explicitly labelled by measure and horizon. Compound options and comparison with reduced-form models remain advanced work.

These are concrete remaining inputs and extensions. The reference code, formulas, symbol definitions and tests above can now be reused in the articles and future institution-specific work.

Part VI. Liquidity and capital components

Reproducible synthetic calculations · reviewed 1 October 2026

These cases connect liquidity eligibility, dated cash, interest earnings, value sensitivity, counterparty exposure and a bounded market-risk capital calculation. The opening bank balance is shared by the liquidity, cash, EVE and NII calculations. The derivative examples are separate, explicitly labelled teaching portfolios.

The calculations are in **research_lab/regulatory_cases.py**; **regulatory_examples()** returns every input, intermediate quantity and result as a serialisable dictionary. Monetary values in Sections 1–3 are **KES million**. Sections 4–7 use **KES**. Rates are decimal fractions; 1 basis point (bp) is 0.0001. Results below are rounded for reading; the code retains floating-point precision.

Abbreviations: LCR = liquidity coverage ratio; NSFR = net stable funding ratio; HQLA = high-quality liquid assets; ASF/RSF = available/required stable funding; SA-CCR = standardised approach to counterparty credit risk; CVA = credit valuation adjustment; BA-CVA = basic approach to CVA capital; FRTB = Fundamental Review of the Trading Book; GIRR = general interest-rate risk. EAD, RC and PFE mean exposure at default, replacement cost and potential future exposure.

What this completes

Review item	Executable coverage	Boundary still open
R16: LCR and NSFR	Eligibility, encumbrance, haircuts, both HQLA composition caps, 75% inflow cap, ASF/RSF and undrawn commitments; reconciliation to one dated cash and equity ledger	Complete institution classification, foreign-currency profile, secured financing unwind, supervisory reporting and real source data
R08: connected stress	Same opening balance produces base/shocked EVE and NII; dated cash response and exact reverse-stress cash threshold	Calibrated deposit behaviour, prepayment, options, full supervisory IRRBB scenarios and joint stochastic balance-sheet simulation
R07: counterparty and CVA	SA-CCR linear single-currency IR netting set, daily cash margin, independent CVA valuation/CS01, wrong-way experiment and reduced BA-CVA capital	Other asset classes and margin structures, full BA-CVA/SA-CVA hedges, market-calibrated exposure simulation and institutional adoption
R15: FRTB	Prescribed GIRR delta sensitivity, tenor weights/correlations, currency aggregation and three correlation scenarios	Other risk classes, applicable vega/curvature, DRC, RRAO, trading-book boundary and total FRTB capital

The Kenyan liquidity factors come from CBK's April 2025 guideline. SA-CCR, BA-CVA and FRTB are pinned **Basel reference implementations**; their appearance here does not establish that the same chapters apply unchanged to every Kenyan institution. KESONIA can supply a benchmark or a curve input. It does not determine instrument eligibility, netting enforceability, collateral availability or a capital rule's institutional perimeter.

1. One opening balance: liquidity and stable funding

1.1 Inputs and classifications

The example bank begins with assets 1,000, liabilities 900 and book equity 100. Its government collateral pledge lasts two years and is unavailable for monetisation. The other securities satisfy the assumed HQLA

quality and operational criteria. “Stable retail” is an aggregate of many qualifying insured KES deposits, not an assumption that an individual KES400 million account is fully insured.

Opening asset	Carrying/market amount	LCR class and haircut	NSFR RSF factor	Required stable funding
Cash	20	Level 1; 0%	0%	0
Available government securities	80	Level 1; 0%	5%	4
Qualifying securities A	100	Level 2A; 15%	15%	15
Qualifying corporate securities B	80	Level 2B; 50%	50%	40
Government securities pledged for two years	20	Excluded: encumbered	100%	20
Performing retail loans due within 30 days	350	Separate cash inflow	50%	175
Performing loans over one year, risk weight above 50%	350	No HQLA inclusion	85%	297.5
Assets	**1,000**			**551.5**

Funding or capital	Amount	ASF factor	Available stable funding	30-day runoff
Capital	100	100%	100	0
Debt with maturity over one year	50	100%	50	0
Stable KES retail deposits	400	95%	380	5%: 20
Less stable KES retail deposits	200	90%	180	10%: 20
Non-financial corporate deposits, below one year	200	50%	100	40%: 80
Financial wholesale funding, below six months	50	0%	0	100%: 50
Funding plus capital	**1,000**		**810**	**170**

An additional irrevocable **credit** commitment to a non-financial corporate has an undrawn amount of 100. Its 30-day draw assumption is 10%, and its NSFR required funding is 5%. A separate contractual collateral call of 10 is payable within 30 days. The performing retail amount due is 350, with eligible LCR inflow 50% × 350 = 175.

These categories use the body of the CBK April 2025 liquidity guideline: LCR §§3.6, 3.16–3.28 and 3.54; NSFR §§3.10–3.15, 3.26–3.36 and Table 3. The case assumes no short securities-financing transactions requiring a 30-day unwind. The function rejects an input profile marked as requiring that unwind.

1.2 Both HQLA caps, applied together

Let L, A and B be eligible Level 1, Level 2A and Level 2B after-haircut amounts, in KES million. Let a and b be the accepted portions of A and B, and $H = L + a + b$ the HQLA total. In this no-unwind profile:

$$0 \le a \le A, \quad 0 \le b \le B, \quad a + b \le 0.40H, \quad b \le 0.15H. \tag{6.1}$$

These constraints imply $H \le L + A + B$, $H \le (L + A)/0.85$ and $H \le L/0.60$. Their feasible maximum is:

$$H = \min\left(L + A + B, \frac{L + A}{0.85}, \frac{L}{0.60}\right). \tag{6.2}$$

Here $L = 100$, $A = 85$, $B = 40$. Thus $H = 166.666667$, with one valid allocation $b = 25$, $a = 41.666667$. The composition caps exclude 58.333333 of the 225 after-haircut pool. Pledged government securities contribute zero.

Let O be weighted outflows, I weighted eligible inflows and I^* recognised inflows. Then:

$$O = 170 + 10 + 10 = 190, \quad I = 175, \quad I^* = \min(I, 0.75O) = 142.5, \tag{6.3}$$

$$N = O - I^* = 47.5, \qquad \text{LCR} = \frac{H}{N} = 3.50877193 = \textbf{350.877193}\%. \tag{6.4}$$

N is 30-day net outflow, in KES million. An HQLA asset cannot also generate an eligible inflow in the input ledger; duplicate identifiers on those sides raise an error. A zero net-outflow denominator is returned as an explicitly undefined ratio.

1.3 Stable funding

ASF is the sum of each funding amount multiplied by its available-stable-funding factor. RSF is the corresponding required-stable-funding sum for assets and commitments. These calculations use **carrying amounts**, rather than the LCR's haircut-reduced amounts:

$$\begin{gathered}\text{ASF} = 810, \quad \text{RSF} = 551.5 + 100(0.05) = 556.5, \\ \text{NSFR} = \frac{810}{556.5} = \textbf{145.552561}\%.\end{gathered} \tag{6.5}$$

The off-balance-sheet commitment requires 5 of stable funding while contributing 10 to the different 30-day liquidity stress. Both originate from the same undrawn commitment of 100.

2. Dated cash, sale losses and reverse stress

2.1 Management response

The ratio calculations use a regulatory horizon. The following scenario adds transaction dates and management actions:

Date	Event	Cash movement	Closing cash	Assets	Liabilities	Equity
1 October	Opening	—	20	1,000	900	100
2 October	Sell government securities carrying 80	+80	100	1,000	900	100
2 October	Sell A carrying 100 for 85	+85	185	985	900	85
2 October	Sell B carrying 10 for 5	+5	190	980	900	80
3 October	Deposit/ wholesale withdrawals	-170	20	810	730	80
3 October	Commitment draw becomes a loan	-10	10	810	730	80
3 October	Post cash collateral, retained as an asset	-10	0	810	730	80
20 October	Collect performing loans	+175	175	810	730	80

Sale prices are deliberately assumed to equal the haircut-adjusted stress prices in this scenario. **A regulatory haircut alone does not create a realised accounting loss.** The assumed actual sale transactions create the 20 loss and reduce equity to 80. Remaining uncollected short loans of 175 stay on the asset ledger; the scenario has not established their impairment or write-off.

Every event satisfies assets = liabilities + equity. The final cash identity is:

$$20 + 170 + 175 - 190 = 175. \tag{6.6}$$

The dated scenario collects 175; the regulatory denominator recognises 142.5 because of its inflow cap. HQLA eligible after composition limits is 166.666667; the targeted sales generate 170. These measure different things, and the ledger makes their reconciliation visible.

Without the sale response, 3 October cash would be **-170** even though the calculated LCR is above 100%. No 20 October receipt can settle a 3 October obligation.

2.2 Reverse cash stress

Let P be proceeds available by 3 October, w a multiplier on the baseline withdrawals of 170, and q an additional collateral call. All cash quantities are KES million:

$$C_{\text{early}} = 20 + P - 170w - 20 - q. \tag{6.7}$$

The fixed 20 is the commitment draw plus the original collateral call. With $q = 0$, the withdrawal multiplier

that exactly exhausts cash is:

$$w^* = \frac{20 + P - 20}{170}. \tag{6.8}$$

Available response	Proceeds P	Withdrawals exhausting cash	w^*	Additional call capacity at baseline $w = 1$	Equity after assumed sales
No sale	0	0	0	Already short 170	100
Targeted sales above	170	170	1.000000	0	80
Sell all unencumbered securities at the same assumed prices	205	205	1.205882	35	45

Equality means cash is exhausted; any greater withdrawal or collateral call creates a deficit. The last response converts the remaining securities to cash but crystallises total losses of 55. It illustrates the cash-versus-capital trade-off. Executability at those prices and same-day settlement still require market and operational evidence.

3. EVE and NII from that same opening balance

EVE is economic value of equity: asset present value minus liability present value. NII is interest income minus interest expense over one year. The amounts below equal the opening LCR/NSFR amounts, with explicit synthetic timing and interest assumptions added.

Instrument	Balance	Final/behavioural life, years	Coupon/base discount rate	First repricing, years	Rate pass-through β
Cash	20	0	0%	0	0
Government	80	3	9%	3	1
A	100	2	10%	2	1
B	80	4	12%	4	1
Pledged government	20	3	9%	3	1
Short retail loans	350	19/365	15%	19/365	1
Long loans	350	3	18%	3	1
Term debt	50	2	11%	2	1
Stable retail deposits	400	2	5%	0	0.5
Less stable retail deposits	200	0.5	6%	0	0.8
Non-financial corporate deposits	200	0.25	7%	0	1
Financial wholesale funding	50	0.1	8%	0	1

The deposit lives and betas are assumptions for examining the mechanism. They are neither estimated customer behaviour nor regulatory runoff factors. Encumbrance removes the pledged bond from available liquidity; its future payments still belong in EVE.

The short-loan life is the 19 calendar days from 1 to 20 October, divided by 365. The stressed liquidity ledger collects 175 of the 350 due and retains 175 as a loan asset; base EVE values the contractual payments, while constant-balance NII replaces maturities.

Cash-flow definitions

For row i, B_i is opening balance; r_i the annual coupon and base annual-effective discount rate; τ_i first repricing time; T_i final life; β_i pass-through; and x the parallel rate shock. Payment dates t_j are annual dates plus any final stub, with $t_0 = 0$. Define the post-reset accrual interval:

$$a_{ij} = \max(0, t_j - \max(t_{j-1}, \tau_i)). \tag{6.9}$$

At each date, the shocked cash flow is:

$$CF_{ij}(x) = B_i r_i (t_j - t_{j-1}) + B_i \beta_i x a_{ij} + B_i \mathbf{1}_{t_j = T_i}. \tag{6.10}$$

1 equals one when its condition holds and zero otherwise. Each principal is repaid exactly once. With $z_i = +1$ for assets and -1 for liabilities:

$$\text{EVE}(x) = \sum_i z_i \sum_j \frac{CF_{ij}(x)}{(1 + r_i + x)^{t_j}}. \tag{6.11}$$

Cash of 20 is included at time zero; equity is the residual and is not discounted as a liability. Existing contracts run off in EVE. NII instead uses a one-year **constant balance** scenario: a maturing position is replaced at the shocked rate for the remaining year.

$$\text{NII}(x) = \sum_i z_i B_i \left[r_i + \beta_i x \max(0, 1 - \tau_i) \right]. \tag{6.12}$$

Output, KES million	Base	+200 bp parallel shock	Change
EVE	99.990652	81.161694	**-18.828958**
One-year NII	88.600000	83.035616	**-5.564384**

The small -0.009348 difference between base EVE and book equity 100 comes from prorated stub coupons discounted on an annual-effective basis. The long integer-year par instruments reconcile exactly. The NII change independently reconciles to:

$$350(0.02)(1 - 19/365) - 0.02[400(0.5) + 200(0.8) + 200 + 50] = -5.564384. \tag{6.13}$$

These are different views of the same opening book: runoff EVE, constant-balance NII and a dated liquidity stress with realised sales. They should not be added together as three independent losses. This section

establishes the numerical bridge for R08; a joint path model, calibrated behavioural assumptions and the full applicable supervisory shock suite remain further work.

4. SA-CCR: netting, collateral and margin

The calculation covers one enforceable interest-rate netting set in one currency, with fixed notionals, linear supervisory deltas and final contractual maturity equal to underlying end time. The three hypothetical derivatives below are measured in KES:

Trade	Notional N_i	Start S_i, years	End/maturity $E_i = M_i$, years	Supervisory delta δ_i	Market value
Short positive-rate-direction trade	5,000,000	0	0.5	+1	150,000
Medium opposite-direction trade	8,000,000	0	3	-1	-250,000
Long positive-rate-direction trade	10,000,000	0	7	+1	500,000

The rule reference is Basel CRE52, published 5 June 2020, effective 1 January 2023. Relevant paragraphs are 52.1–52.2, 52.10–52.23, 52.34, 52.48–52.53 and 52.57.

With 250 business days per year, supervisory duration SD_i, in years, and maturity factors MF_i, dimensionless, are:

$$SD_i = \mathsf{max}\left(\frac{10}{250}, \frac{e^{-0.05S_i} - e^{-0.05E_i}}{0.05}\right), \qquad MF_i^U = \sqrt{\mathsf{min}(\mathsf{max}(M_i, 10/250), 1)}, \tag{6.14}$$

$$MF_i^M = 1.5\sqrt{\mathsf{MPOR}/250}, \qquad d_i = N_i SD_i, \quad D_i = d_i \delta_i MF_i. \tag{6.15}$$

MPOR is the margin period of risk in business days. The daily bilateral cash-margin example uses 10 days and assumes no longer floor is triggered by illiquidity, size or disputes. Calculated duration and unmargined remaining maturity have **separate** ten-business-day floors.

Let D_1, D_2, D_3 be summed effective notionals in end-time buckets below one year, one through five years, and over five years:

$$EN = \sqrt{D_1^2 + D_2^2 + D_3^2 + 1.4D_1D_2 + 1.4D_2D_3 + 0.6D_1D_3}, \qquad A_{\mathsf{CCR}} = 0.005EN. \tag{6.16}$$

EN and A_{CCR} are currency amounts under the supervisory annual time convention. Let V be net market value, C net eligible cash collateral received, TH threshold, MTA minimum transfer amount, and $NICA$ net independent collateral amount, all in KES:

$$RC_U = \mathsf{max}(V - C, 0), \qquad RC_M = \mathsf{max}(V - C, TH + MTA - NICA, 0), \tag{6.17}$$

$$m = \min\left(1,\ 0.05 + 0.95\exp\left[\frac{V-C}{1.9A_{\text{CCR}}}\right]\right), \quad PFE = mA_{\text{CCR}}, \quad EAD_{\text{raw}} = 1.4(RC + PFE). \tag{6.18}$$

When the add-on is zero, the code directly sets PFE to zero and avoids division. Margined EAD is capped by the same netting set's unmargined EAD. The sample collateral is same-currency cash, so the same C applies in both cap calculations. A non-cash implementation would need separate horizon-appropriate collateral haircuts.

Quantity, KES unless stated	Unmargined	Daily cash margin
V	400,000	400,000
C	0	450,000
$TH, MTA, NICA$	Not used	0; 10,000; 50,000
Bucket 1 effective notional	1,745,852.863	740,702.639
Bucket 2	-22,286,723.772	-6,686,017.132
Bucket 3	59,062,382.056	17,718,714.617
Aggregate add-on	231,984.736	69,694.139
Replacement cost	400,000	0
Multiplier m	1	0.701236
PFE	231,984.736	48,872.048
EAD	**884,778.630**	**68,420.867**

The daily margin consists of 400,000 variation margin plus 50,000 independent collateral. Its unmargined cap is 291,690.805 and is not binding. Calculating the three unmargined trades separately gives total EAD 1,388,963.115, so recognised netting reduces it by 504,184.485. That benefit requires the legal and operational netting assumptions to hold.

EAD is an exposure measure. It becomes a credit capital input only through the relevant counterparty risk-weighting framework; it is not itself a capital charge.

5. CVA valuation, spread sensitivity and wrong-way risk

This is a separate unilateral valuation model with deterministic discounting and synthetic exposure. Let recovery $R = 0.4$, loss fraction $g = 1 - R = 0.6$, annual spread $s = 0.02$, flat intensity $\lambda = s/g$, and continuously compounded discount rate $r = 0.09$. The spread/intensity relation is a model assumption.

For interval j, t_{j-1} and t_j are year boundaries, q_j the marginal default probability, EE_j exposure conditional on default in that interval, and DF_j the discount factor:

$$q_j = e^{-\lambda t_{j-1}} - e^{-\lambda t_j}, \quad DF_j = e^{-rt_j}, \qquad \text{CVA} = \sum_j g\, EE_j DF_j q_j. \tag{6.19}$$

CVA is reported as a positive loss in KES:

t_j, years	EE_j, KES	q_j	Discount factor	CVA contribution, KES
1	600,000	0.032783900	0.913931185	10,786.402
2	900,000	0.031709115	0.835270211	14,302.267
3	400,000	0.030669567	0.763379494	5,619.004
Total				**30,707.674**

Here CS01 is **KES change in CVA loss per one-basis-point spread increase**, with exposures and recovery held fixed. It is not the FRTB sensitivity divided by the decimal bump:

$$\text{CS01} = 10^{-4}\frac{\partial \text{CVA}}{\partial s} = 10^{-4}\sum_j EE_j DF_j\left[t_j e^{-\lambda t_j} - t_{j-1}e^{-\lambda t_{j-1}}\right]. \tag{6.20}$$

The analytic CS01 is **146.736899**. A central ±1 bp calculation gives 146.736903; a one-sided +1 bp valuation change is 146.703901. Halving the central bump reduces derivative error approximately quadratically.

A hypothetical protection sensitivity of 88.042139 would offset 60% and leave 58.694759 KES/bp. This is a sensitivity allocation only. Executable hedge instrument valuation, basis risk, cost, eligibility and capital recognition require separate inputs.

Preserving default marginals when testing wrong-way risk

Use two persistent states, each with probability 0.5. The calm state has spread 0.4% and exposure scale 500,000; the stress state has spread 3.6% and exposure scale 1,500,000. Both use exposure proportions $1, 0.8, 0.4$ at years $1, 2, 3$, recovery 40% and discount rate 9%.

The joint calculation averages state-specific $gEE_jDF_jq_j$. The independence comparator uses the **same** weighted marginal q_j and the same weighted mean EE_j, then multiplies those means. Consequently the measured difference comes from dependence, rather than changing the marginal default curve.

Result	KES
CVA with positive exposure/default dependence	49,411.850
Independence CVA, identical marginal inputs	35,438.254
Wrong-way increment	**13,973.596**

This valuation experiment explains a mechanism; it does not estimate a real counterparty's wrong-way risk.

6. Reduced BA-CVA capital: a distinct calculation

Basel MAR50, published 8 July 2020 and effective 1 January 2023, supplies the reduced basic-approach formulas in §§50.13–50.16. This version recognises no hedge benefit.

For counterparty c, RW_c is its sector/credit-quality risk weight; netting set n has EAD in KES and effective maturity M_n in years. For the non-IMM profile, where IMM means internal models method:

$$DF_n = \frac{1 - e^{-0.05M_n}}{0.05M_n}, \qquad SCVA_c = \frac{RW_c}{1.4}\sum_n M_n EAD_n DF_n. \tag{6.21}$$

With supervisory parameter $\rho = 0.5$, discount scalar $d = 0.65$, and capital K_{BA}:

$$K_{\text{reduced}} = \sqrt{\left(\rho\sum_c SCVA_c\right)^2 + (1-\rho^2)\sum_c SCVA_c^2}, \quad K_{\text{BA}} = dK_{\text{reduced}}, \quad RWA = 12.5K_{\text{BA}}. \tag{6.22}$$

RWA denotes risk-weighted assets. It and capital are both currency amounts; they are different quantities.

The example uses one hypothetical investment-grade industrial counterparty, $RW = 3\%$, with a single three-year, single-payment IR derivative: notional 3,000,000, market value 100,000, supervisory delta +1 and no collateral. The preceding SA-CCR implementation gives EAD 198,502.650. Effective maturity is 3 for this deliberately single-payment example; final maturity must not substitute automatically for effective maturity in a general portfolio.

Quantity	Result
Supervisory discount factor	0.9286134905
Maturity-adjusted exposure $MEADDF$	KES552,996.716
Stand-alone $SCVA$	KES11,849.930
Reduced BA-CVA capital	**KES7,702.454**
Associated RWA	**KES96,280.678**

EAD already contains the SA-CCR factor 1.4; the BA-CVA formula divides it out once. Neither the CVA valuation of Section 5 nor its illustrative 60% sensitivity offset is substituted into this capital formula. Full BA-CVA with qualifying hedges and SA-CVA remain separate extensions.

7. FRTB: a fully worked GIRR delta component

The pinned reference is Basel MAR21, publication **23 March 2026**, effective 1 January 2023. The source supplies netting/aggregation (§21.4), correlation scenarios (§21.6), sensitivity (§21.19), weights (§21.42), tenor correlations (§21.46) and currency correlation (§21.50).

This portfolio has one risk-free zero curve per currency. Its zero-coupon positions are synthetic, with all notionals already converted to the common KES reporting unit and FX held fixed. A negative notional is a short position.

For signed notional n, continuous zero rate r and tenor T in years:

$$\begin{gathered}\mathrm{PV} = ne^{-rT},\\ s_T = \frac{\mathrm{PV}(r+0.0001) - \mathrm{PV}(r)}{0.0001} = \frac{ne^{-rT}(e^{-0.0001T}-1)}{0.0001}.\end{gathered} \tag{6.23}$$

s_T is KES per unit **decimal** rate; multiplying it by 0.0001 gives the one-basis-point PV movement. The code uses the prescribed finite +1 bp shift. The analytic derivative $-Tne^{-rT}$ is retained as a separate numerical cross-check.

Currency/tenor	Signed reporting notional, KES	Zero rate	s_T, KES per decimal	Risk weight	Weighted sensitivity WS_T, KES
KES 1 year	1,000,000	9%	-913,885.490	1.6%	-14,622.168
KES 5 years	-700,000	10%	2,122,326.683	1.1%	23,345.594
USD 2 years	900,000	4%	-1,661,443.274	1.3%	-21,598.763
USD 10 years	-400,000	5%	2,424,909.982	1.1%	26,674.010

The weights are unreduced. The optional division by $\sqrt{2}$ for specified currencies/domestic reporting currency is not elected.

Net same-factor sensitivities before weighting: $WS_k = RW_k s_k$, where k indexes a currency/tenor factor and RW_k is its decimal risk weight. For two different tenors T_k, T_l on the same curve:

$$\rho_{kl} = \max\left(e^{-0.03|T_k - T_l|/\min(T_k, T_l)}, 0.4\right). \tag{6.24}$$

Both 1-versus-5 and 2-versus-10 year pairs here have medium correlation $e^{-0.12} = 0.886920437$. For currency bucket b:

$$K_b = \sqrt{\max\left(0, \sum_{k\in b} WS_k^2 + 2\sum_{k<l;\ k,l\in b} \rho_{kl} WS_k WS_l\right)}, \quad S_b = \sum_{k\in b} WS_k. \tag{6.25}$$

Across currency buckets the medium correlation is $\gamma = 0.5$:

$$K_{\text{GIRR},\Delta} = \sqrt{\sum_b K_b^2 + 2\sum_{b<c} \gamma_{bc} S_b S_c}. \tag{6.26}$$

If that aggregate radicand is negative, the implementation recomputes it using S_b bounded to $[-K_b, K_b]$. Here that alternative is unnecessary.

For each correlation c, the high scenario uses $\min(1.25c, 1)$, and the low scenario uses $\max(2c-1, 0.75c)$. Apply the same scenario to tenor and currency correlations:

Scenario	KES bucket K_b	USD bucket K_b	Across-currency correlation	Aggregate GIRR delta, KES
Medium	12,381.460	12,492.165	0.500	18,805.007
High	8,723.426	5,075.247	0.625	12,537.871
Low	15,182.324	16,921.887	0.375	**23,453.323**

The low-correlation scenario dominates because long/short tenor offsets become less effective. This is the maximum for the **GIRR yield-curve delta component**. A complete portfolio selects its governing scenario after the required portfolio aggregation.

The calculation does not produce total FRTB capital. Inflation, cross-currency basis, other curves, FX risk and credit-spread risk are outside this example. Vega/curvature for optional instruments, default risk charge (DRC), residual risk add-on (RRAO), trading-book classification and local adoption require their own coverage. The hypothetical zero-coupon positions have no embedded option, but that fact does not settle those components for a real book.

8. Reproduction, tests and next evidence

From the repository root:

```
python -m unittest discover -s tests -p test_regulatory_cases.py -v

python -c "import json; from research_lab.regulatory_cases import regulatory_examples;
print(json.dumps(regulatory_examples(), indent=2, allow_nan=False))"
```

The module uses the Python standard library and writes nothing on import. All **30 tests passed** on 1 October 2026. The test suite checks:

- simultaneous HQLA cap feasibility, encumbrance, inflow double counting, undefined denominators and ratio scale invariance;
- ASF/RSF reconciliation and the cash/equity identity at every dated event;
- the shared opening balance, zero-shock EVE/NII identity, independent NII repricing arithmetic, one principal repayment per cash-flow schedule and exact reverse-stress boundary/breach;
- SA-CCR hand calculations, exact offsets, maturity/duration floors, margin cap, collateral multiplier bounds and currency scaling;
- a closed-form CVA benchmark, convergence of the numerical CS01, unchanged marginal probabilities in the wrong-way comparator, and BA-CVA alpha/scalar/diversification arithmetic;
- tenor correlations, exact risk-factor netting, two-factor hand calculations, scale/sign/order invariance and independent finite-shift/analytic sensitivity checks.

Completion evidence is therefore mathematical and computational for these stated profiles. Institution data, parameter estimation, market quotations, executable collateral operations and applicability approval are the next evidence layers. These calculations make those requirements concrete without treating the unfinished wider frameworks as completed.

Part VII. Accounting and connected finance

Edition: 1 October 2026. Reproduce with `python -m research_lab.run_examples`. Machine-readable output is in `research_lab/results/checked_examples.json`, under `accounting` and `connected`. All amounts below are synthetic KES. Designation eligibility, actual fair values, legal transfer and distribution permissions require evidence from the relevant owner.

1. Fair-value hedge and subsequent amortisation

The example elects an amortised-cost asset with two remaining annual receipts: 50 and 1,050. Its carrying amount before the benchmark change is 1,000. A benchmark-only revaluation from 5% to 6% gives:

$$\begin{aligned} V_{\text{adjusted}} &= 50/1.06 + 1050/1.06^2 = 981.666073, \\ L_{\text{item}} &= 1000 - V_{\text{adjusted}} = 18.333927, \\ G_{\text{derivative}} &= 18.000000, \\ G_{\text{ineffectiveness}} &= 18.000000 - 18.333927 = -0.333927. \end{aligned} \tag{7.1}$$

Here V_{adjusted} is adjusted carrying amount; L_{item} is the hedged-item loss; $G_{\text{derivative}}$ is the supplied derivative gain; and $G_{\text{ineffectiveness}}$ is net hedge ineffectiveness. All quantities are in KES.

The asset loss and derivative gain enter profit or loss separately. The journal record uses positive debits and negative credits; each event balances to zero. Derivative settlement exchanges an 18 derivative asset for 18 cash. On discontinuation, the asset adjustment remains in its carrying amount and is amortised using the recalculated 6% effective interest rate (EIR). The two-year roll-forward reaches zero after receipt of the contractual 1,100 total. The calculation measures the adjustment and amortisation; it assumes the hedge was properly designated and qualified.

Event	Debit	Credit
Asset benchmark remeasurement	Hedge loss 18.333927	Asset carrying adjustment 18.333927
Derivative remeasurement	Derivative asset 18	Hedge gain 18
Derivative settlement	Cash 18	Derivative asset 18
Subsequent effective-interest recognition	Asset carrying amount: opening balance × 6%	Interest income: same amount
Subsequent coupon/principal receipt	Cash: 50 then 1,050	Asset carrying amount: same amounts

2. Cash-flow hedge: cumulative reserve and cash reconciliation

The designated instrument's supplied cumulative gains are 12, 17 and 20. The corresponding hypothetical hedged-item changes are -10, -16 and -19. The fixture keeps the same designation throughout this measurement sequence.

Date	Derivative cumulative gain	Opposing item change	Cumulative OCI reserve	Cumulative ineffectiveness gain	Period ineffectiveness gain
31 March 2027	12	-10	10	2	2
30 June 2027	17	-16	16	1	-1
30 September 2027	20	-19	19	1	0

OCI means other comprehensive income. The reserve is capped using the smaller absolute cumulative change, retaining the instrument's sign. The period journal uses differences between cumulative balances. Applying a fresh period-by-period minimum would give the wrong answer when prior ineffectiveness reverses.

At settlement, the borrower pays contractual interest of 100 and receives derivative cash of 20. Reclassifying 19 from the reserve reduces interest expense to 81. The separately recognised ineffectiveness gain of 1 gives total profit-or-loss cost of 80, matching net cash paid of 80. The reserve and derivative finish at zero. Underhedging is also tested: an instrument gain of 8 against an item loss of 10 creates a reserve of 8 without inventing an extra accounting loss of 2.

3. Rebalancing and discontinuation

The separate control example records designation D1 with 1,000 hedged units and 1,000 hedge units. An unchanged risk objective and completed effectiveness review permit a prospective version with 900 hedge units. Original quantities and prior ineffectiveness stay in the history. A changed objective fails this rebalancing gate. The fixture does not infer later fair values merely by changing the ratio; a new valuation is required.

At June discontinuation, the accumulated reserve is 16. If the forecast cash flows remain expected, it is retained for the appropriate later recognition. If they are no longer expected, 16 is reclassified immediately. These two branches are calculated independently of the completed September hedge; they are alternative histories, not extra journals added to it.

The measurement principles are checked against IFRS 9 §§6.4.1, 6.5.5–6.5.12 through the New Zealand standard setter's published IFRS 9 text. This is access to the accounting text, not a claim that New Zealand jurisdiction determines a Kenyan institution's obligations. Instrument eligibility, economic relationship, credit-risk dominance, documentation, hypothetical-derivative construction and approval remain inputs to a qualified accounting implementation.

4. A common borrower, originator, SPV and investor cash case

The borrower reuses Underwrite for Collection's on-time 60-day case: opening cash 250,000; operating receipts 1,000,000; operating payments 820,000; loan draw 500,000; upfront fee 10,000; final debt payment 515,000. Closing borrower cash is 405,000 and financing cost is 25,000. Every transfer has a payer, recipient, date and unique event identifier.

To demonstrate a claim transfer, the bank originates the 500,000 loan and an illustrative SPV buys it at par on the same date. These are new interface assumptions. The originator retains its 10,000 fee; the SPV receives the loan's 15,000 interest. An assumed 75/15/10 asset funding split supplies 375,000, 75,000 and 50,000. Class C supplies an additional 10,000 to fund the closing reserve. Thus the 510,000 cash raised funds both purchase and reserve. This small fixture does not replace Mwendo's complete waterfall or establish legal/accounting sale treatment.

On 30 November, the SPV's 515,000 collection plus 10,000 reserve pays, in order:

Use of cash	KES
Servicing	2,000
Class A coupon	6,000
Class B coupon	1,500
Class A principal	375,000

Use of cash	KES
Class B principal	75,000
Class C asset principal	50,000
Final reserve release and residual to C	15,500
Total	525,000

Class C receives 65,500 on its total 60,000 investment. Only 5,500 is residual profit before investor tax; returned capital and released reserve are not profit. The SPV closes with no assets, claims or cash in this successful termination case. Failure, trigger, delayed recovery and persistent arrears cases remain in Mwendo's canonical engine and its versioned evidence snapshot.

The originator incurs 8,000 operating cost and an assumed 600 tax on its 2,000 fee contribution. Its 1,400 profit supports 600 retention and a permitted 800 parent remittance on 15 December. The loan interest belongs to the SPV after the assumed transfer; adding it to the bank's fee again would overstate originator income. Borrower cost reconciles exactly to originator fee plus SPV loan interest.

5. Accepted refund, received cash and owner endpoints

The separate insurance interface assumes a 1,200 premium, twelve-month service period and elected pro-rata cancellation after three months. An accepted 900 refund on 31 March creates a right/payable, with zero SPV cash at that date. Settlement on 15 April extinguishes that right, transfers 900 cash and reduces a 1,000 loan to 100. Counting acceptance and settlement as two recoveries would fail this reconciliation. The illustrative service allocation of 300 is not asserted to be IFRS 17 insurance revenue. Contract groups, remaining/incurred coverage liabilities, onerous measurement, claims and insurer journals remain with the insurance accounting owner.

The owner example separately reconciles 1,590,000 reported benefit less replacement labour 300,000, maintenance 180,000 and required working capital 120,000 to earning power of 990,000. Today's 200,000 unrestricted cash less 80,000 near-term claims and 100,000 buffer permits only 20,000 distribution. Neither endpoint is silently relabelled as the other.

At the bank parent, 100 opening cash cannot meet a 300 obligation on 1 December. The 800 remittance on 15 December arrives later. A temporary 200 bridge is needed; after receiving the remittance and repaying that bridge, parent cash is 600. Costs and tax already deducted in the bank are not charged again to the net remittance.

6. Typed handoff and SokoIntel candidate

`EconomicQuantity` requires a finite Decimal value, unit, currency, legal entity, claim, period, scenario and source version. Addition requires all dimensions to match. Missing information fails validation; it is not converted to zero. The transfer ledger rejects overdrafts, out-of-order new events, unknown entities and changed payloads under reused identifiers; an exact duplicate has no second cash effect.

These are executable candidate contracts and acceptance fixtures for SokoIntel. The guest project was not edited and no SokoIntel runtime was certified. RegTech preserves a dated snapshot of the owning-project evidence so that the examples run independently of another user's Downloads folders.

Symbol or field	Meaning and unit
KES	Kenya shillings; all money in this chapter

Symbol or field	Meaning and unit
EIR	Annual effective interest rate; 0.06 in the fair-value amortisation fixture
FV	Fair value, in currency
OCI / P&L	Other comprehensive income / profit or loss; accounting presentation categories
D1	Designation version identifier; not a discount factor
`derivative`, `reserve`, `ineffectiveness`	Cumulative instrument value change, effective OCI amount and residual gain; KES with explicit signs
`signed_debits`	Positive debit and negative credit amounts; sum zero for each journal
`event_id`	Unique transfer identifier whose payload is immutable
75/15/10	Percentage allocation of the 500,000 asset purchase; excludes separately funded reserve
`cash_residual`	Sum of entity cash after a transfer less the sum before it; KES, required zero

Implementation: `accounting_cases.py` and `connected_cases.py`; tests in the corresponding test modules. Full contract-group IFRS 17, empirical loss models, live cash execution and institution-approved distribution rules remain separate completion requirements.

Part VIII. Implementation and remaining work

Reconciled 1 October 2026. The audit parsed all **71 existing Markdown and HTML documents** outside caches/vendor folders: 47 Markdown and 24 HTML. The baseline inventory records hashes and text coverage; the baseline audit records what was missing before this update. Parsing a document establishes coverage of its text, not independent validation of each inherited assertion.

1. Main conclusion

KESONIA supplies the reference **within** CBK's revised Risk-Based Credit Pricing Model. Borrower risk, lending costs and shareholder return determine the institution's approved customer premium; FTP explains internal funding/liquidity economics. These are complementary layers. The earlier wording about a transition "from RBCPM to KESONIA" should not be used. Part 1 and the active bibliography now state this relationship explicitly. CBK issued framework, §§4–5.

The September review correctly identified work beyond compounding, but its delivery status became stale as code was added in RegTech, Mwendo and Underwrite. This revision supplies missing reproducible **reference implementations** and retains a separate list of unfinished research, institutional and integration work. No broad R-item below is claimed to be a completed bank deployment.

2. Read and reproduce

Reading route	Implementation / result key
Original worked examples	`benchmark.py`, `curves.py`, `pricing.py`, `scenarios.py`; original result keys
FTP, deposits and existing STP	`operations.py`; `operations`
Dated accrual, correction and workflow extensions	`lifecycle_cases.py`; `lifecycle`
OAS, priced hedges, collateral and structural credit	`option_cases.py`; `options`
Liquidity, exposure, CVA and trading-book components	`regulatory_cases.py`; `regulatory`
Hedge accounting and connected financing	`accounting_cases.py`, `connected_cases.py`; `accounting`, `connected`
Symbols and units	Original glossary plus links to extension-specific definitions
SokoIntel candidate map	Product candidates and acceptance fixtures; no guest implementation claimed
Screenshot synthesis: ALM, credit loss and workbench boundaries	Proactive ALM/ALCO, IRRBB/NMD/FTP framing, Treasury control proposal, credit-loss tail illustration and project/export boundaries

From the repository root, run:

```
python -m unittest discover -s tests -v

python -m research_lab.run_examples

python publications/build_publications.py

python publications/build_worked_examples_pdf.py
```

The calculation code uses the Python standard library. HTML and PDF publication commands need the documented rendering dependencies. The deterministic result file is checked_examples.json. Source and data hashes connect that output to its implementation. Tests check independent identities, limiting cases, replay, accounting balance and cash conservation, alongside invalid-input failures.

3. Eighteen-item reconciliation

"Reference implemented" means the stated bounded fixture runs and has tests. "Partial research" identifies executable progress with substantive model work still open. "Owning project" prevents duplicate methods and records the current external evidence.

ID	Delivery status and executable evidence	Work still required
R01 Curves	**Reference implemented:** annual deposit/par bootstrap, interpolation, separate projection/discount curves, repricing and sensitivity convergence; options tree additionally fits all monthly synthetic discounts	Dated executable Kenyan quotes, instrument calendars, stale-quote/version controls and multi-instrument market calibration
R02 FTP lifecycle	**Reference implemented:** fixed, amortising, floating and revolving dated funding, commitment costs, ex-ante/ex-post bridge and consolidated elimination in `operations.py`	Complete volume/rate/basis/loss attribution, actual servicing ledger, institutional optionality/liquidity policies
R03 Deposits	**Partial research:** segmented asymmetric lagged regression, temporal holdout, retention/migration and constrained ladder	Permissioned data, robust uncertainty, feedback into dynamic reinvestment/withdrawal decisions and empirical stress validation
R04 Prepayment/OAS	**Reference implemented:** monthly curve-fitted short-rate tree, amortising pool, CPR/SMM, assumed rate-responsive prepayment, optimal-exercise comparison and numerical OAS solve in `option_cases.py`	Hull–White/Black–Karasinski alternative calibration, market volatility data, real borrower behaviour and measure/identification validation
R05 Hedge choice	**Reference implemented:** priced two-tenor swaps, finite portfolio search, basis/twist stress, slope/curvature, fees and initial-cash limits; original PV01 illustration retained	Executable prices, liquidity/counterparty/product constraints and dynamic rebalancing
R06 Collateral	**Reference implemented:** dated/intraday bilateral versus cleared cash, netting, threshold/MTA, haircuts, IM/VM and funding reconciliation	Actual CSA/CCP terms, disputes/default closeout, full instrument calendar roll, operational margin feed
R07 SA-CCR/CVA	**Scoped components implemented:** single-currency interest-rate netting/margin exposure, CVA valuation/CS01 and wrong-way comparison, reduced BA-CVA illustration	Other asset classes, full hedged BA/SA-CVA, actual netting/eligibility, local rule mapping; valuation hedging is not automatic capital relief
R08 Bank path	**Reference implemented:** original three-period cash/equity roll; new common-balance liquidity, NII/EVE and reverse-stress scenario	One fully dynamic institution-calibrated accounting/ALM/capital/distribution simulation and actual returns
R09 STP	**Reference implemented:** original durable proposal-to-reconciliation flow; partial fill/settlement, unknown acknowledgement, cancellation/amendment, correction approval/replay extensions	Unify reference flows with real vendor schemas, authentication, acknowledgements, capacity/recovery, exception-service standards and operational approval

ID	Delivery status and executable evidence	Work still required
R10 Mwendo SPV	**Owning-project reference exists:** October 1 canonical case covers funded closing reserve, priority payments, triggers, arrears and delayed refund/recovery; RegTech freezes provenance and adds a small handoff fixture	Live pool/servicer evidence, enforceable transaction terms, distribution/credit/rate stress calibration; reuse Mwendo engine
R11 Insurance	**Cash interface implemented:** accepted refund, settlement and lender/SPV allocation have distinct dates, rights and conservation tests	Full IFRS 17 group measurement, incurred claims, insurer and lender/SPV journals and approved handoff from accounting owner
R12 Structural credit	**Partial research:** Merton terminal-debt/equity valuation and inverse synthetic calibration	Compound-option payoff/exercise model, coupon/barrier extensions, observable data and reduced-form comparison
R13 Bayesian/statistical	**Owning-project methods recognised:** credit timing/cure, hierarchy/shrinkage and insurance relativities retain their canonical owners	Permissioned empirical, temporal, subgroup, ablation and decision-value experiments; not duplicated as KESONIA functionality
R14 Borrower-to-owner	**Reference implemented:** common borrower/originator/SPV/investor cash ledger, tax/retention, distinct earning power and distribution cash, delayed parent remittance; typed quantity checks	Institution/group-specific capital, tax, approvals and cross-project production schemas; connected fixture remains deliberately small
R15 FRTB	**Component implemented:** scoped standardised GIRR delta calculation with tenor/currency aggregation and correlation scenarios	Full relevant risk classes, vega/curvature, DRC/RRAO; IMA liquidity-horizon/stress ES, RFET, P&L attribution/backtesting and supervisory approval
R16 LCR/NSFR	**Reference implemented:** explicit eligibility, encumbrance, HQLA caps, runoff/inflow cap, ASF/RSF and undrawn commitment; linked to available cash	Complete bank classifications/returns, actual collateral monetisation and regulatory reporting reconciliation
R17 Hedge accounting	**Reference implemented:** fair-value hedge journals, recalculated-EIR amortisation, cumulative cash-flow reserve, ineffectiveness, settlement/reclassification, discontinuation alternatives and prospective designation control	Instrument valuation after rebalancing, real designation/effectiveness/credit-risk assessment, accounting approval and GL integration
R18 Kenya scope	**Verified source map plus research profiles:** issued CBK framework, historical fixing/index check, local liquidity guidance; source/version and evidence gates in fixtures	Actual entity/product contracts, authoritative unresolved day mapping, local available instruments/quotes, approved rules and reporting endpoints

4. Cross-project acceptance cases

The boundary document, section 13, reconciles Cases A–F and actions XB01–XB10. Its source map preserves September 28 reading anchors. The new snapshot at cross_project_reference_2026-10-01.json records the current owning-project results and hashes.

Mwendo's existing 28-test verification and Underwrite's saved borrower/decision cases are recognised as **external reference evidence**. They were read, not independently recertified by RegTech's test run. Neither a snapshot nor this repository's candidate map proves that SokoIntel has implemented those contracts.

5. Historical material and remaining delivery priorities

The September 23 pricing formulations, September 25 treasury casebook/software/FRTB reviews, original gap/enhancement notes, and screenshot-driven cross-project material informed this work. Historical review landing pages now point here. The old bibliography is preserved before targeted corrections to active claims about RBCPM, SHAP, benchmark-reform accounting and mandated cryptography. Bibliographic metadata outside the checked source register remains inherited material.

Remaining work has three concrete groups:

1. **Further code and research:** quote/calibration governance, full FTP attribution and behavioural uncertainty; unified dynamic bank/operational adapters; fuller capital approaches; post-rebalancing valuation and compound options. These are genuine unfinished developments, not merely missing approvals.
2. **Owning-project work:** insurance accounting, Mwendo transaction evidence and empirical Bayesian/credit experiments. Existing methodology and simulations are reused.
3. **Institutional evidence and execution:** permissioned histories, market quotes, signed terms, local classifications, approved calendars, vendor environments and supervisory/accounting acceptance.

This completes the stated new reference examples while preserving those broader requirements. The series and future SokoIntel programme remain larger than this review pack.

6. Screenshot-derived scope and export additions

The 4 October screenshot synthesis adds a reading and modelling checklist to the existing bounded examples: NII and EVE must be reported as different IRRBB views; deposit beta/pass-through, repricing lag, runoff and migration require segment-specific evidence and stress; "free funds" is a conditional spread outcome; and ALCO risk appetite, model-risk challenge and Treasury execution authority must remain visible. These topics sharpen R03/R08/R16 but do not turn the current examples into a calibrated institution's balance-sheet model.

The screenshot's expected-versus-unexpected-loss bridge is now stated with an explicit two-loan dependence illustration. It distinguishes approximate one-period $PD \times LGD \times EAD$, IFRS 9 discounted cash-shortfall ECL, internally defined economic capital, and rule-based regulatory capital. The illustration is a conceptual review fixture in the note, not new checked Python or a validated capital method.

SokoIntel now has two additional candidate exports: (a) a versioned ALCO scenario/decision packet with NII/EVE, liquidity/collateral, binding constraints, sensitivity and approval state; and (b) a portfolio expected-loss/tail-risk bridge that identifies its method and dependence assumptions and labels ECL, economic-capital and regulatory-capital outputs separately. Both remain candidate requirements for product-owner mapping. The workbench may explain or carry approval records but must not select risk appetite, approve a capital method, or execute a hedge or funding transaction.

Part IX. SokoIntel candidates

1 October 2026. All introduced material can inform the workbench. Product scope should distinguish a calculation users can inspect, a specialist analytical module, an acceptance fixture and an integration contract. This is a candidate map saved in RegTech. SokoIntel's guest repository remains unchanged; adopted requirements are not presented as implemented software.

1. Candidate modules and release order

Priority	Candidate capability	Reusable reference and acceptance condition
Core	Borrower/owner cash and purpose analysis	Underwrite/Mwendo owning cases; `connected_cases.py`: every cash transfer has payer, recipient and date; earning power and permitted distributions remain distinct
Core	Customer pricing and lending contribution	`pricing.py`, `operations.py`: benchmark plus margin reconciles to funding, cost, expected principal loss, fees and capital-return money without double counting
Core	Typed quantities, point-in-time evidence and scenarios	`connected_cases.py`, `lifecycle_cases.py`: incompatible entity/unit/claim/date quantities fail; unavailable evidence preserves its freshness state
Specialist treasury	Curves, FTP, deposits and liquidity	`curves.py`, `operations.py`, `regulatory_cases.py`: instruments reprice, internal transfers eliminate, cash and prudential ratios are separately reconciled
Specialist treasury	ALCO scenario and constrained-policy evidence export	`regulatory_cases.py` and the screenshot synthesis: dated NII/EVE, liquidity/collateral, behavioural-deposit assumptions, binding limits, uncertainty, stress results, version lineage and approval state in one review packet; actions remain proposals until authorized
Specialist risk	Credit expected-loss to portfolio-tail bridge	Underwrite owns borrower PD/LGD/EAD and event outcomes; candidate adapter presents the portfolio loss distribution and dependence assumptions, with IFRS 9 ECL, economic capital and applicable regulatory capital in separately labelled views; no generic formula is presented as booked ECL or required capital
Specialist finance	SPV and protection/refund adapters	Mwendo canonical waterfall plus RegTech cash fixtures: funded reserves, persistent arrears and distinct acceptance/settlement dates
Specialist accounting	Hedge-accounting explanation and journal preview	`accounting_cases.py`: journal balance, cumulative reserve, discontinuation and cash/P&L reconciliation; posting authority stays with Finance/GL

Priority	Candidate capability	Reusable reference and acceptance condition
Advanced analytics	OAS, optionality and constrained hedge comparisons	`option_cases.py`: curve fit, valuation-measure assumptions, numerical residuals and funding demands visible
Advanced regulatory	SA-CCR, CVA and FRTB components	`regulatory_cases.py`: explicit method/version/asset-class perimeter; component output cannot be labelled total capital
Research	Bayesian uncertainty, structural/compound credit and behavioural estimation	Canonical credit/relativities owners; experiments promoted only after temporal validation and material decision value
Integration	Approval, order, confirmation, booking and settlement contracts	`operations.py`, `lifecycle_cases.py`: replay, partial execution and exceptions as adapter acceptance tests; live execution belongs to connected operational systems

2. Existing specification and new evidence

The September cross-project review already identified adopted SokoIntel requirements FR099–FR132. This update supplies executable evidence that can be mapped to those requirements during a product-owner review. It does not issue new FR numbers or claim exact requirement closure without inspecting the current controlled specification and implementation together.

Each candidate should carry: intended user and decision; source and model version; legal entity and claim; currency, units and horizon; input evidence status; result uncertainty; calculation/reconciliation tests; supported limitations; and approval owner where the output affects a contract, journal or instruction. Treasury exports should additionally identify NII horizon and EVE discount assumptions, NMD behaviour vintage, the exact binding risk appetite/limit and whether an action is approved, proposed or rejected. Risk exports should show dependence, confidence/risk measure, expected loss, quantile/shortfall measure, accounting basis and prudential method as distinct fields.

The workbench can explain how KESONIA, the approved customer premium, FTP, expected loss and capital-return amounts fit together. It should allow separate views of the same dated financing chain without merging borrower, lender, SPV, insurer and shareholder balances.

3. Practical promotion gates

1. Reproduce the case and its failure scenarios in the workbench's own test environment.
2. Bind inputs to versioned, typed evidence with explicit missing-data handling.
3. Provide plain-language result explanations and units beside each output.
4. Validate the module for the intended population, instrument and decision.
5. Integrate through documented, authorised APIs when execution or accounting is required.

An illustration is useful as an acceptance fixture before it becomes a product feature. Specialist capital and trading-book modules should be selected for users who need those methods; they are not prerequisites for an MSME cash-flow analysis screen.

See the implementation register for current status and the cross-project boundaries for ownership. The

new screenshot synthesis records these candidates and their limits. They are review/export requirements only: the maintained RegTech repository does not implement a sequential RL controller or an autonomous treasury execution service, and no guest SokoIntel files were changed.